

NKPC HSA Bayesian State-Space Estimation: Output gap, BN filter

June 23, 2026

1 Data and Transformations

Raw data are read from `data/raw/`. Processed model-ready data are written to `data/processed/`. The HSA competition aggregator convention is

$$N_t^{model} = \frac{100 \log(N_t^{level}) - \overline{100 \log(N^{level})}}{10}.$$

Thus N_t^{model} , $\hat{N}_t$, and $\bar{N}_t$ are centered and measured in ten-log-point units. If a run metadata file records another `n_transform`, interpret the corresponding coefficients using that metadata.

2 Model Specifications

The CES benchmark is

$$\pi_t = \alpha \pi_{t-1} + (1 - \alpha) E_t \pi_{t+1} + \kappa x_t + e_t.$$

The dynamic HSA specification is

$$\pi_t = \alpha \pi_{t-1} + (1 - \alpha) E_t \pi_{t+1} + \kappa x_t - \theta \hat{N}_t + e_t.$$

The steady HSA specification is

$$\pi_t = \alpha \pi_{t-1} + (1 - \alpha) E_t \pi_{t+1} + (\kappa_0 + \delta \bar{N}_t) x_t + e_t.$$

The both-effects HSA specification is

$$\pi_t = \alpha \pi_{t-1} + (1 - \alpha) E_t \pi_{t+1} + (\kappa_0 + \delta \bar{N}_t) x_t - (\theta_0 + \gamma \bar{N}_t) \hat{N}_t + e_t.$$

The maintained HSA decomposition is

$$N_t = \hat{N}_t + \bar{N}_t, \quad \hat{N}_t = \rho_1 \hat{N}_{t-1} + \rho_2 \hat{N}_{t-2} + u_t, \quad \bar{N}_t = n + \bar{N}_{t-1} + \epsilon_t.$$

3 Unit and Covariance Conventions

Priors for κ , κ_0 , and δ are specified in physical units. Legacy Gibbs samplers internally use `KAPPA_SCALE=100` when the design matrix contains $x_t/100$. All posterior draws, tables, figures, SDDR calculations, and model-comparison summaries are reported in physical units. The default dynamic-HSA shock covariance allows only (e_t, ζ_t) correlation; diagonal and historical full covariance modes are available only through explicit run metadata. Because the default N transform is centered and divided by 10, the reported δ , θ , θ_0 , and γ coefficients are already effects for a ten-log-point movement in the corresponding N component. κ_0 and θ_0 are intercepts at average $\bar{N}_t$.

4 Coefficient Summary

Table 1 reports posterior means and 95% credible intervals for key parameters across all four models (baseline prior, full sample). Cells show *mean* [*ci_2.5*, *ci_97.5*].

Table 1: Posterior means and 95% credible intervals by model

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.672 [0.556, 0.785]	0.657 [0.541, 0.771]	0.636 [0.511, 0.758]	0.604 [0.484, 0.725]
kappa	0.059 [-0.025, 0.144]	NaN	0.084 [-0.009, 0.181]	NaN
kappa_0	NaN	0.046 [-0.043, 0.133]	NaN	0.088 [-0.009, 0.179]
delta	NaN	-0.023 [-0.056, 0.010]	NaN	-0.022 [-0.051, 0.015]
theta	NaN	NaN	0.117 [-0.356, 0.646]	NaN
theta_0	NaN	NaN	NaN	0.076 [-0.283, 0.538]
gamma	NaN	NaN	NaN	0.001 [-0.036, 0.033]
rho_1	NaN	0.736 [0.364, 1.047]	0.912 [0.664, 1.118]	0.988 [0.756, 1.216]
rho_2	NaN	-0.028 [-0.319, 0.242]	0.017 [-0.177, 0.235]	-0.042 [-0.257, 0.179]
phi_1	0.915 [0.848, 0.982]	0.915 [0.853, 0.978]	0.916 [0.855, 0.979]	0.914 [0.850, 0.977]
lambda_ez	0.010 [-0.220, 0.237]	-0.003 [-0.224, 0.225]	NaN	-0.022 [-0.250, 0.196]
n	NaN	-0.021 [-0.079, 0.039]	-0.025 [-0.079, 0.025]	-0.030 [-0.100, 0.039]

5 Result Blocks

Results are organized by condition. Each section fixes a data specification, prior specification, and sample period, then reports the coefficient table, prior/posterior overlay by model, Savage-Dickey density ratios, time-varying κ_t path, and model-comparison output for that condition.

6 output_gap_bn: prior=baseline, period=full

Condition. data=output_gap_bn, prior=baseline, period=full, constraint=unrestricted (unrestricted).

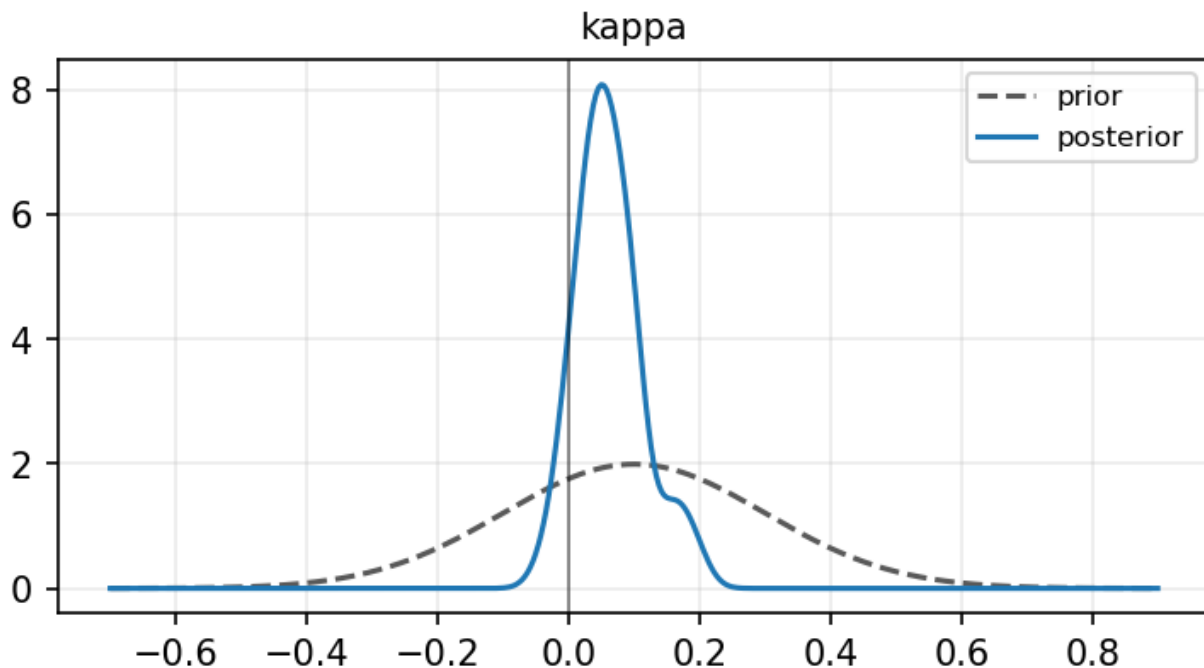
6.1 Coefficient Table

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.669 [0.544, 0.784]	0.655 [0.534, 0.754]	0.642 [0.541, 0.730]	0.662 [0.563, 0.758]
kappa	0.062 [-0.006, 0.176]	NaN	0.081 [-0.004, 0.150]	NaN
kappa_0	NaN	0.136 [-0.243, 0.429]	NaN	0.248 [-0.039, 0.547]
delta	NaN	-0.000 [-0.002, 0.002]	NaN	-0.001 [-0.002, 0.001]
theta	NaN	NaN	0.009 [0.001, 0.014]	NaN
theta_0	NaN	NaN	NaN	0.101 [-0.175, 0.409]
gamma	NaN	NaN	NaN	-0.000 [-0.002, 0.001]
rho_1	NaN	1.802 [1.700, 1.893]	1.862 [1.794, 1.913]	1.191 [1.003, 1.388]
rho_2	NaN	-0.806 [-0.900, -0.703]	-0.866 [-0.917, -0.797]	-0.269 [-0.470, -0.067]
phi_1	0.919 [0.838, 0.970]	0.918 [0.867, 0.974]	0.907 [0.861, 0.953]	0.916 [0.872, 0.976]
lambda_ez	-0.009 [-0.205, 0.221]	0.001 [-0.161, 0.161]	NaN	-0.013 [-0.195, 0.272]
n	NaN	-0.057 [-0.200, 0.059]	-0.026 [-0.150, 0.106]	-0.090 [-0.181, 0.037]

6.2 Prior vs Posterior by Model

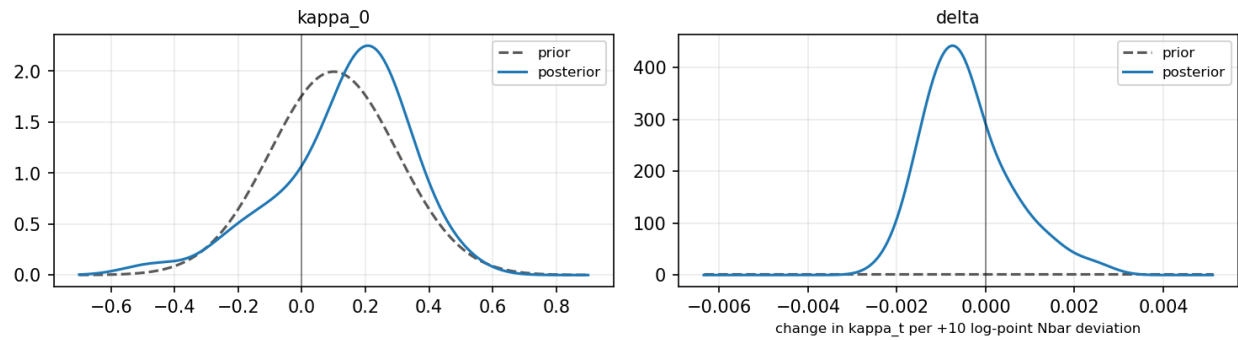
ces

Prior vs posterior — ces



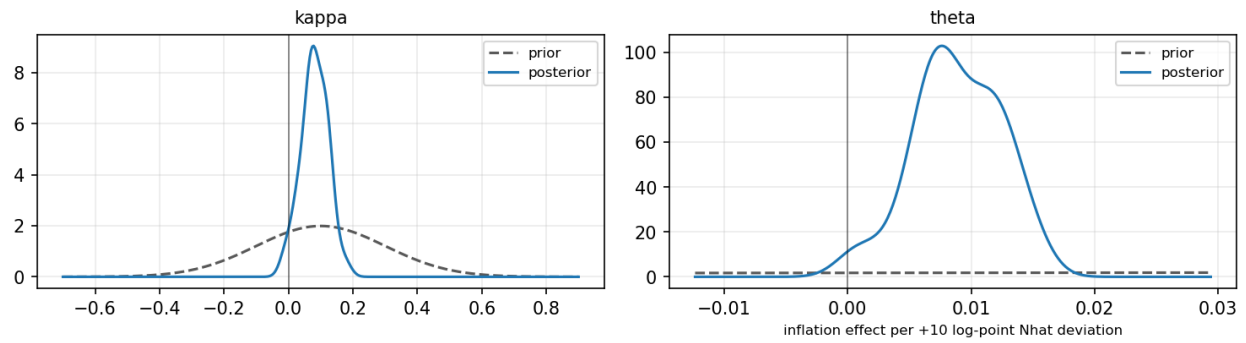
hsa_steady

Prior vs posterior — hsa_steady



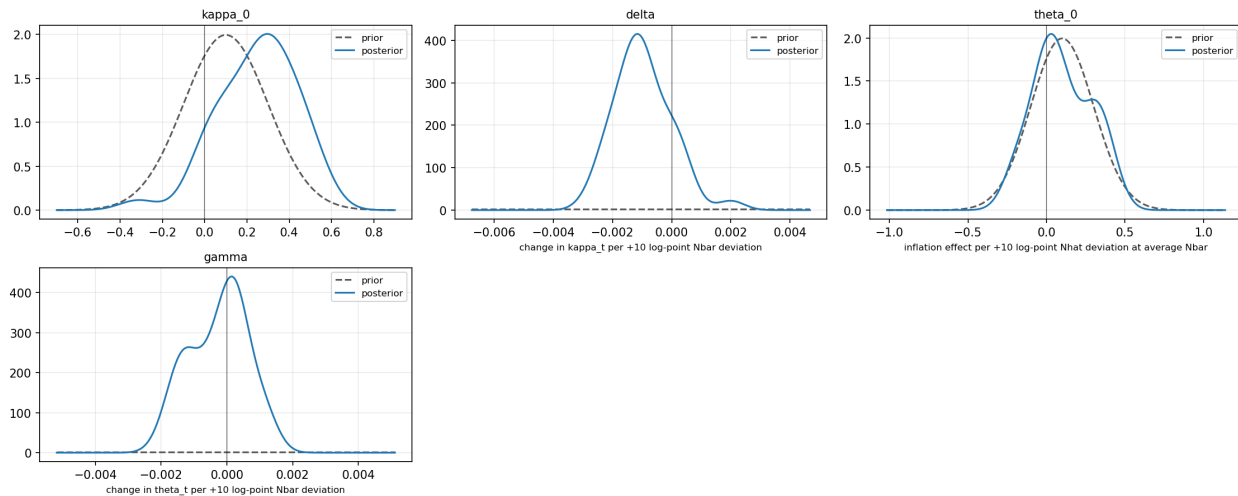
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

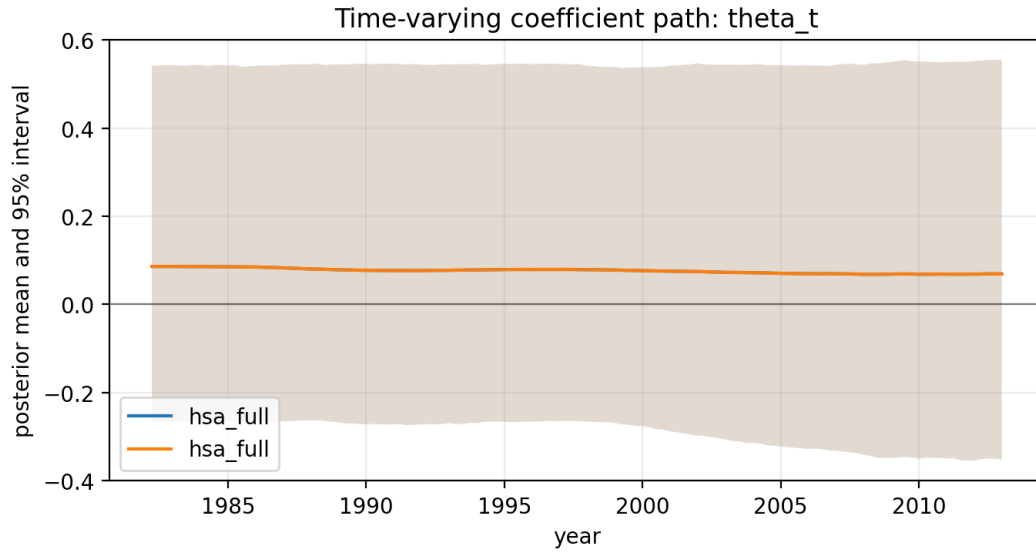
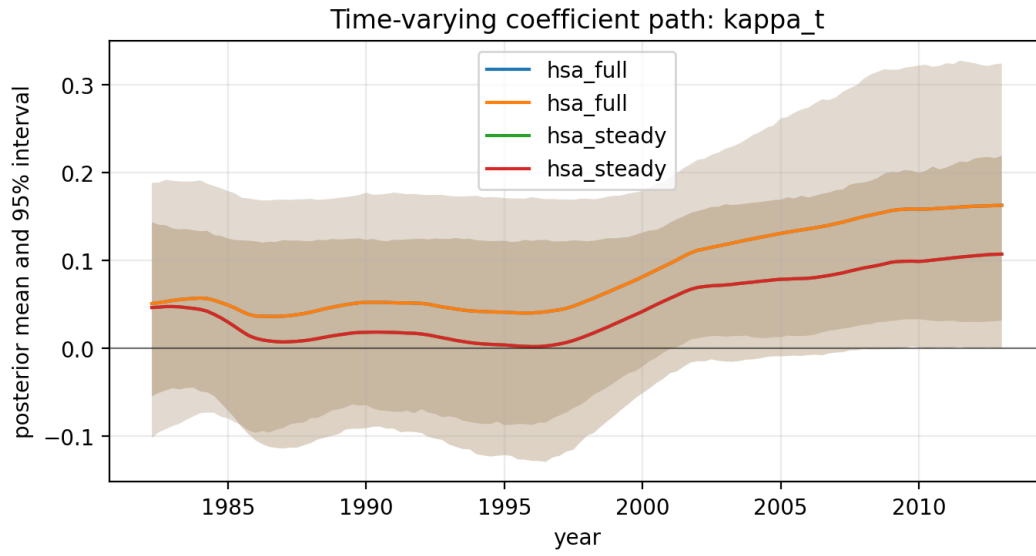
Prior vs posterior — hsa_full



6.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	2.0246
ces	kappa=0	0.1000	0.2000	2.0246
hsa_dynamic	kappa=0	0.1000	0.2000	1.1406
hsa_dynamic	theta=0	0.1000	0.2000	0.2352
hsa_dynamic	kappa=0	0.1000	0.2000	1.1406
hsa_dynamic	theta=0	0.1000	0.2000	0.2352
hsa_full	kappa_0=0	0.1000	0.2000	0.9794
hsa_full	delta=0	0.0000	0.0200	0.4393
hsa_full	theta_0=0	0.1000	0.2000	0.0800
hsa_full	gamma=0	0.0000	0.0200	1.2421
hsa_full	kappa_0=0	0.1000	0.2000	0.9794
hsa_full	delta=0	0.0000	0.0200	0.4393
hsa_full	theta_0=0	0.1000	0.2000	0.0800
hsa_full	gamma=0	0.0000	0.0200	1.2421
hsa_steady	kappa_0=0	0.1000	0.2000	2.8901
hsa_steady	delta=0	0.0000	0.0200	0.4690
hsa_steady	kappa_0=0	0.1000	0.2000	2.8901
hsa_steady	delta=0	0.0000	0.0200	0.4690

6.4 Time-Varying Kappa Path



6.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-120.7533	0.6396	-245.3547	1.0000	NaN	NaN	NaN
ces	NaN	NaN	-245.3547	1.0000	NaN	NaN	NaN
hsa_dynamic	-121.9815	0.6441	NaN	NaN	NaN	0.2352	NaN
hsa_dynamic	NaN	NaN	NaN	NaN	NaN	0.2352	NaN
hsa_full	-123.1413	0.6504	NaN	NaN	0.4393	NaN	1.2421
hsa_full	-123.1413	0.6504	NaN	NaN	0.4393	NaN	1.2421
hsa_steady	-118.8999	0.6300	-306.7120	0.0000	0.4690	NaN	NaN
hsa_steady	NaN	NaN	-306.7120	0.0000	0.4690	NaN	NaN

7 output_gap_bn: prior=baseline, period=full, constraint=kappa $i=0$

Condition. data=output_gap_bn, prior=baseline, period=full, constraint=restricted_kappa (kappa $i=0$).

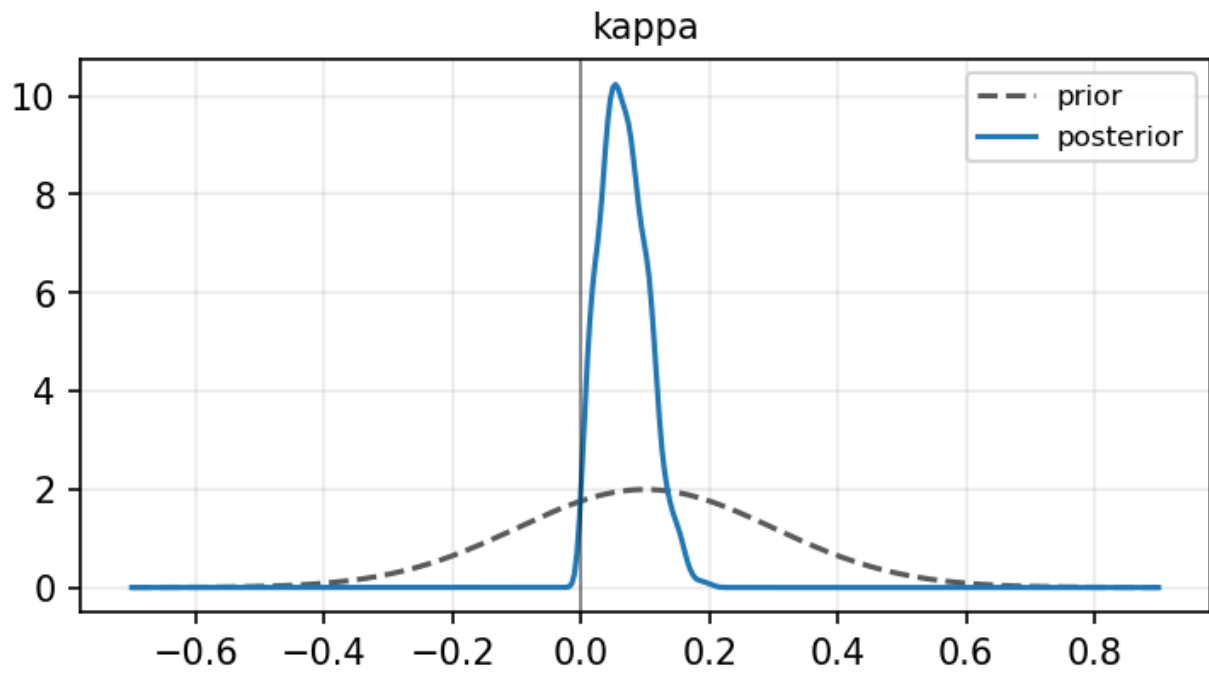
7.1 Coefficient Table

	ces	hsa_steady	hsa_dynamic	hsa_full
parameter				
alpha	0.674 [0.552, 0.788]	0.657 [0.541, 0.771]	0.633 [0.514, 0.754]	0.604 [0.484, 0.725]
kappa	0.067 [0.007, 0.147]	NaN	0.091 [0.013, 0.187]	NaN
kappa_0	NaN	0.046 [-0.043, 0.133]	NaN	0.088 [-0.009, 0.179]
delta	NaN	-0.023 [-0.056, 0.010]	NaN	-0.022 [-0.051, 0.015]
theta	NaN	NaN	0.176 [-0.335, 0.658]	NaN
theta_0	NaN	NaN	NaN	0.076 [-0.283, 0.538]
gamma	NaN	NaN	NaN	0.001 [-0.036, 0.033]
rho_1	NaN	0.736 [0.364, 1.047]	0.916 [0.669, 1.121]	0.988 [0.756, 1.216]
rho_2	NaN	-0.028 [-0.319, 0.242]	0.016 [-0.176, 0.235]	-0.042 [-0.257, 0.179]
phi_1	0.916 [0.851, 0.982]	0.915 [0.853, 0.978]	0.915 [0.852, 0.978]	0.914 [0.850, 0.977]
lambda_ez	0.004 [-0.230, 0.227]	-0.003 [-0.224, 0.225]	NaN	-0.022 [-0.250, 0.196]
n	NaN	-0.021 [-0.079, 0.039]	-0.022 [-0.076, 0.027]	-0.030 [-0.100, 0.039]

7.2 Prior vs Posterior by Model

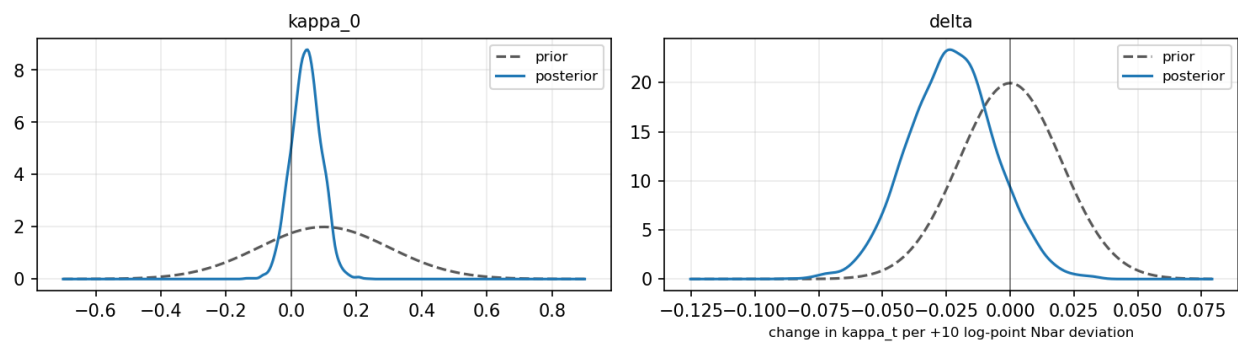
ces

Prior vs posterior — ces



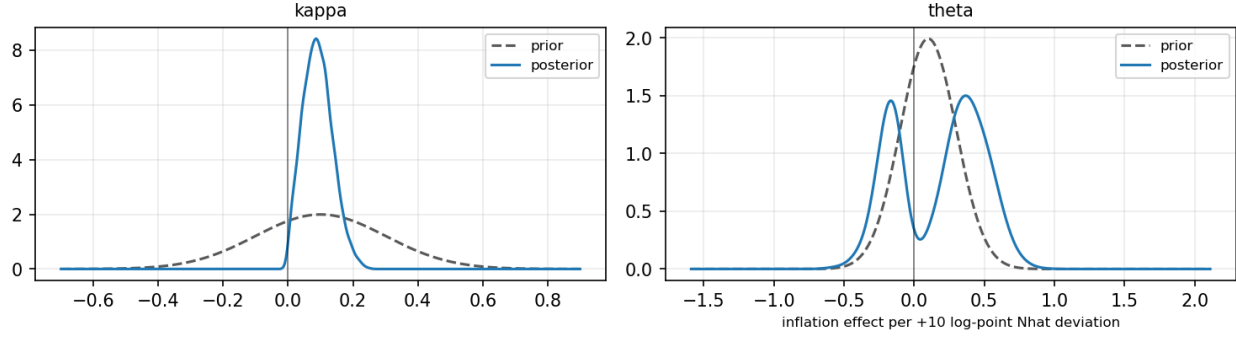
hsa_steady

Prior vs posterior — hsa_steady



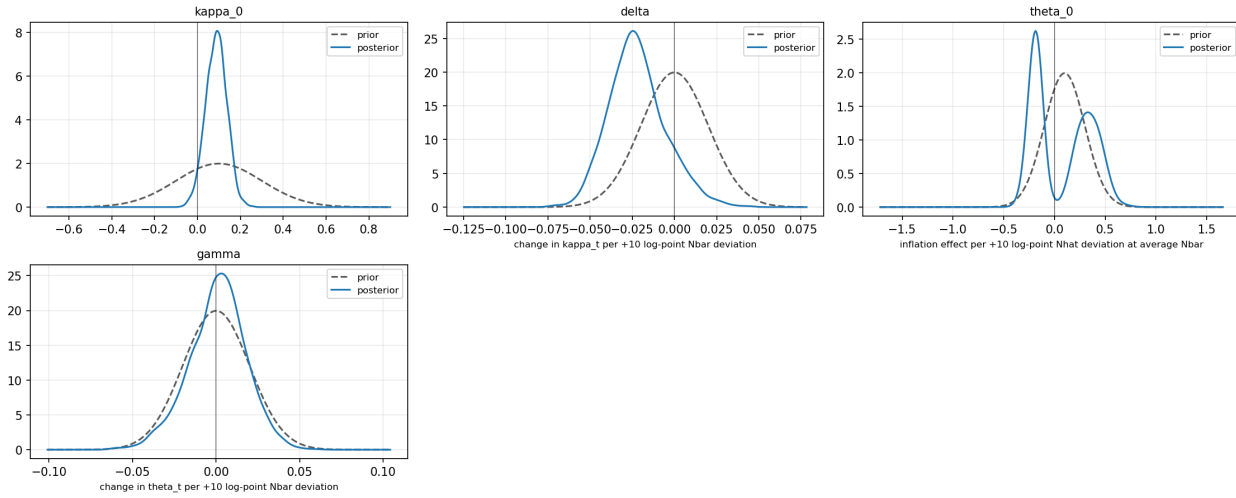
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

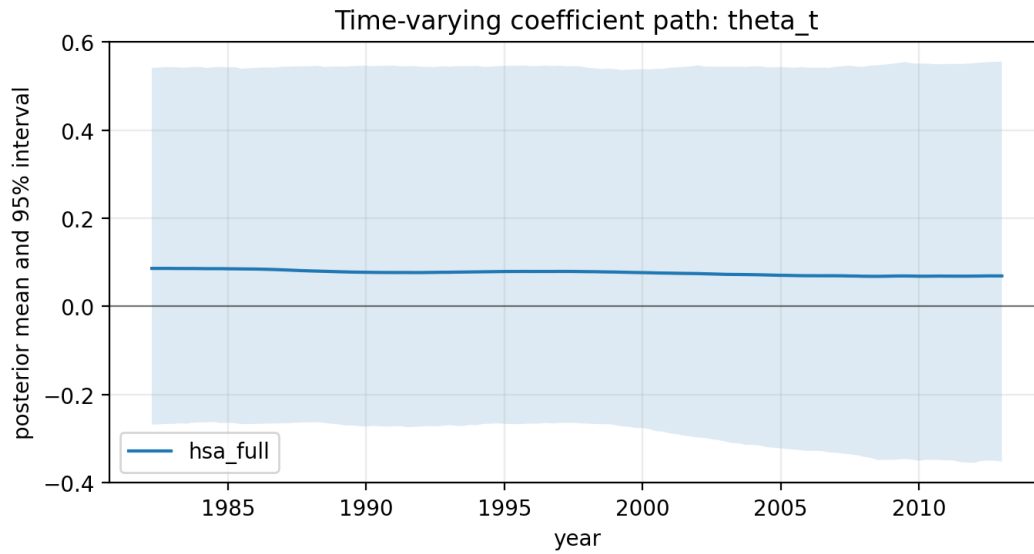
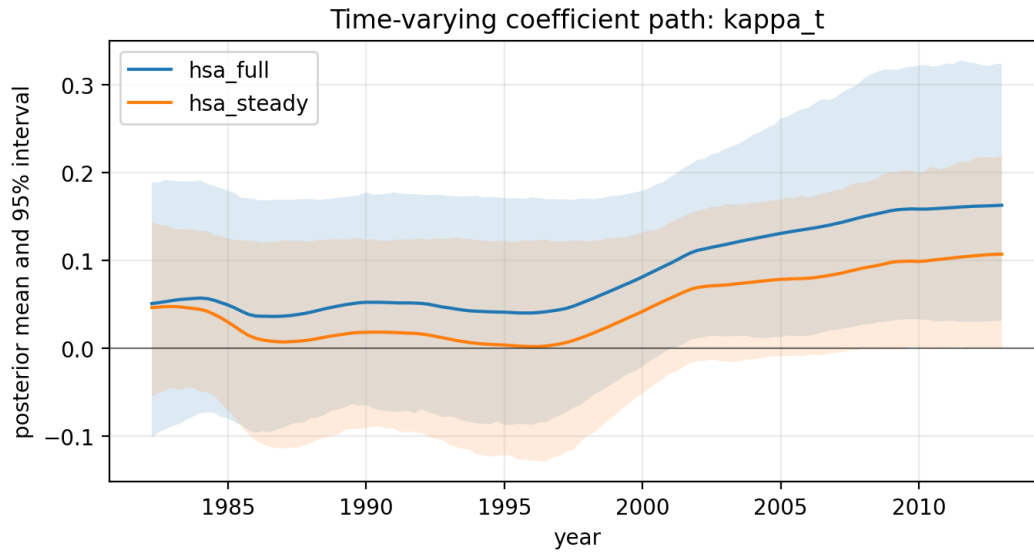
Prior vs posterior — hsa_full



7.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	1.1343
hsa_dynamic	kappa=0	0.1000	0.2000	0.5746
hsa_dynamic	theta=0	0.1000	0.2000	0.1935
hsa_full	kappa_0=0	0.1000	0.2000	0.9794
hsa_full	delta=0	0.0000	0.0200	0.4393
hsa_full	theta_0=0	0.1000	0.2000	0.0800
hsa_full	gamma=0	0.0000	0.0200	1.2421
hsa_steady	kappa_0=0	0.1000	0.2000	2.8901
hsa_steady	delta=0	0.0000	0.0200	0.4690

7.4 Time-Varying Kappa Path



7.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-120.7733	0.6397	-245.3581	1.0000	NaN	NaN	NaN
hsa_dynamic	-120.1361	0.6354	NaN	NaN	NaN	0.1935	NaN
hsa_full	-123.1413	0.6504	NaN	NaN	0.4393	NaN	1.2421
hsa_steady	-118.8999	0.6300	-306.7120	0.0000	0.4690	NaN	NaN

8 output_gap_bn: prior=baseline, period=pre_2008

Condition. data=output_gap_bn, prior=baseline, period=pre_2008, constraint=unrestricted (unrestricted).

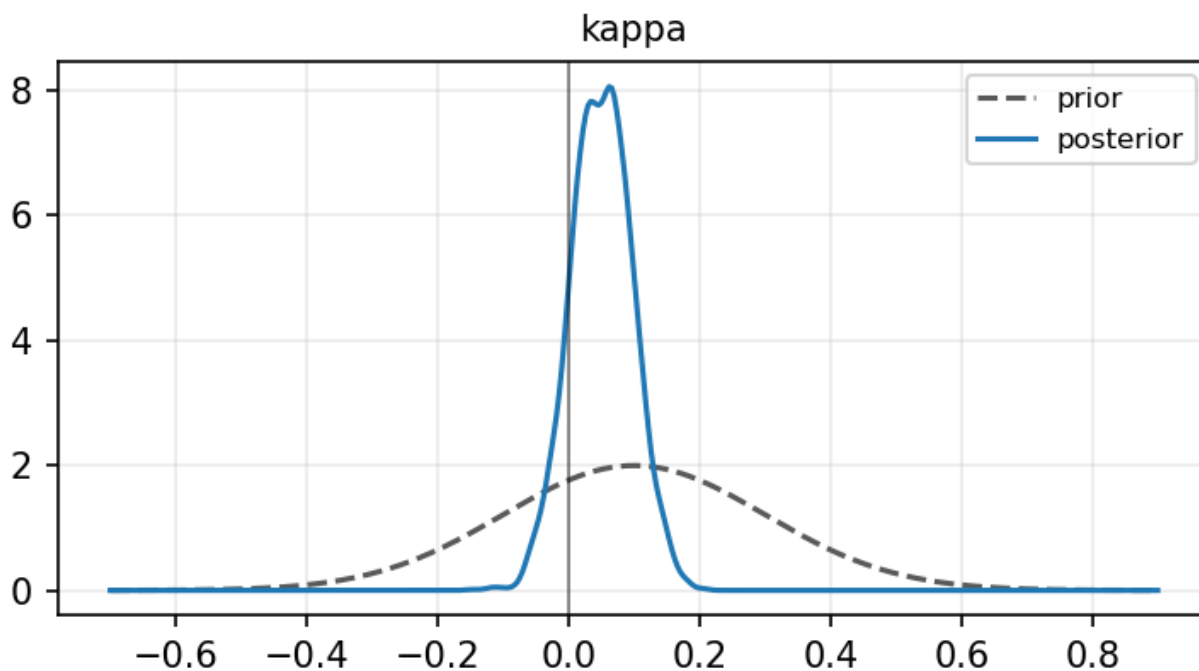
8.1 Coefficient Table

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.708 [0.571, 0.842]	0.724 [0.614, 0.850]	0.624 [0.489, 0.766]	0.626 [0.477, 0.778]
kappa	0.050 [-0.042, 0.142]	NaN	0.069 [-0.021, 0.162]	NaN
kappa_0	NaN	0.120 [-0.158, 0.492]	NaN	0.059 [-0.045, 0.163]
delta	NaN	-0.000 [-0.002, 0.001]	NaN	-0.009 [-0.044, 0.025]
theta	NaN	NaN	0.384 [-0.285, 0.726]	NaN
theta_0	NaN	NaN	NaN	0.323 [0.086, 0.583]
gamma	NaN	NaN	NaN	-0.005 [-0.043, 0.034]
rho_1	NaN	1.814 [1.705, 1.898]	0.900 [0.664, 1.120]	0.955 [0.720, 1.187]
rho_2	NaN	-0.821 [-0.905, -0.710]	0.008 [-0.202, 0.219]	-0.039 [-0.263, 0.197]
phi_1	0.907 [0.834, 0.981]	0.905 [0.842, 0.993]	0.908 [0.838, 0.979]	0.907 [0.834, 0.979]
lambda_ez	-0.118 [-0.339, 0.103]	-0.127 [-0.292, 0.031]	NaN	-0.135 [-0.351, 0.072]
n	NaN	-0.032 [-0.168, 0.108]	-0.008 [-0.067, 0.043]	-0.001 [-0.059, 0.060]

8.2 Prior vs Posterior by Model

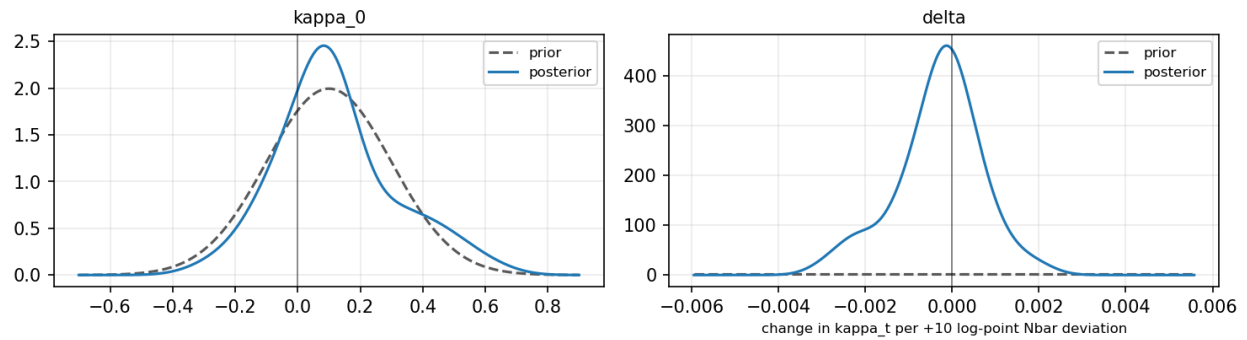
ces

Prior vs posterior — ces



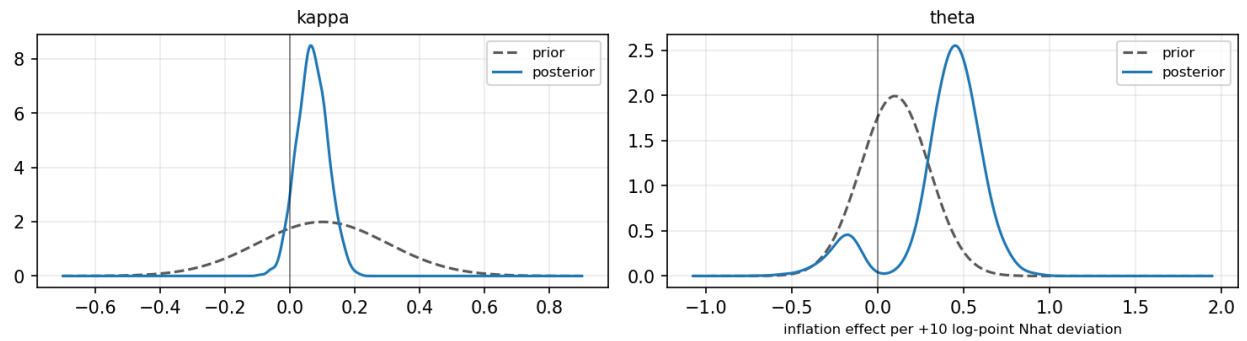
hsa_steady

Prior vs posterior — hsa_steady



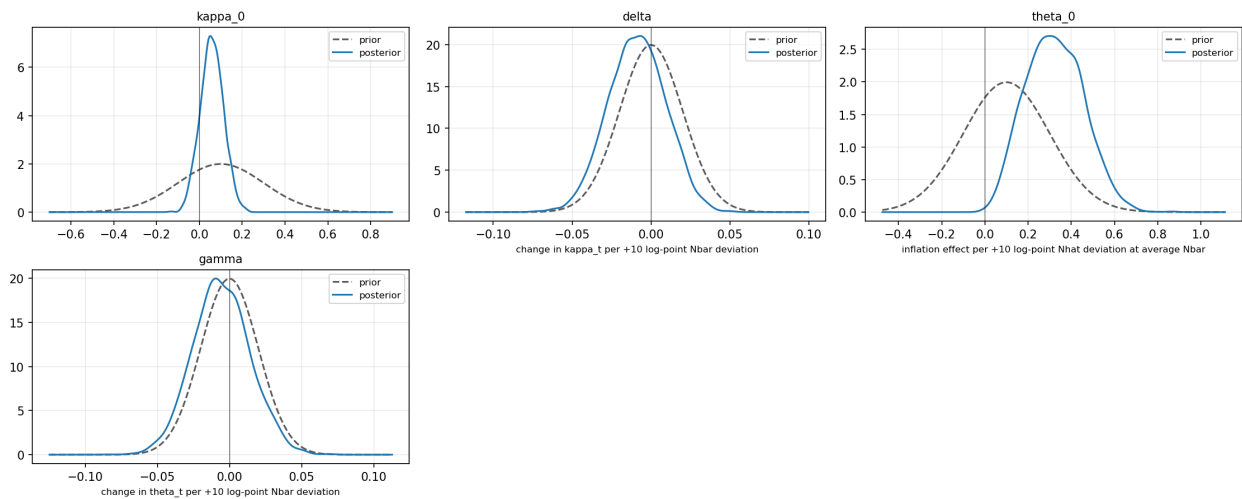
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

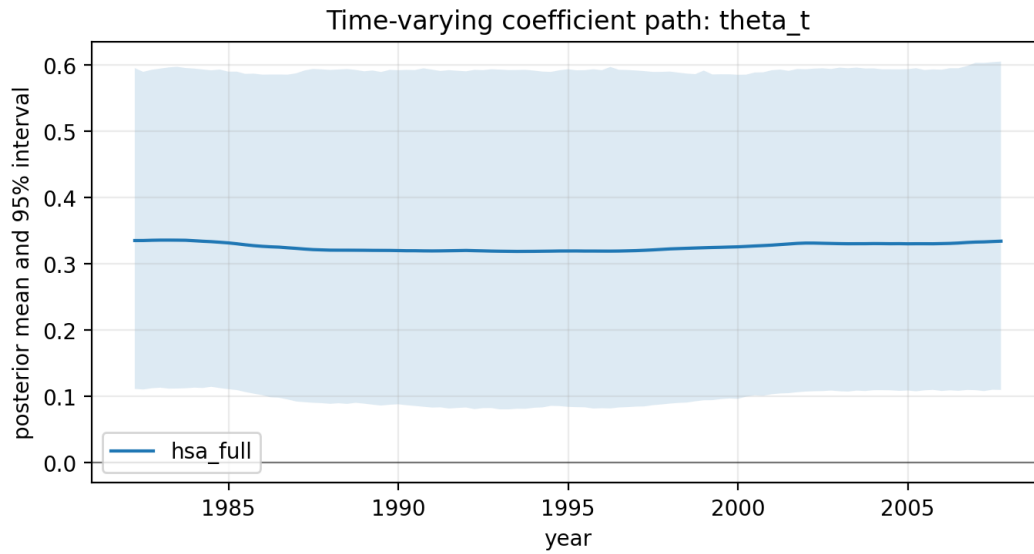
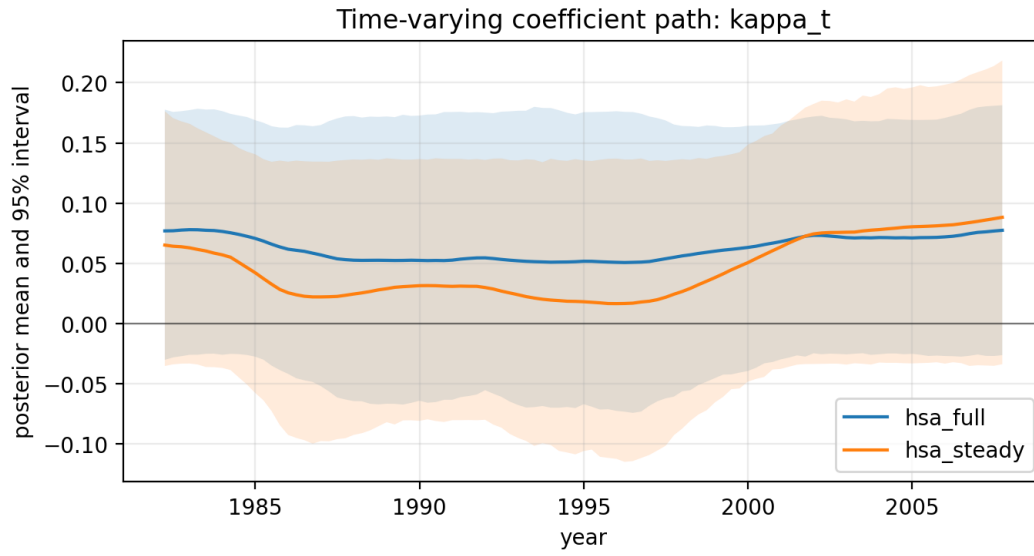
Prior vs posterior — hsa_full



8.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	2.7661
hsa_dynamic	kappa=0	0.1000	0.2000	1.7255
hsa_dynamic	theta=0	0.1000	0.2000	0.0245
hsa_full	kappa_0=0	0.1000	0.2000	2.2381
hsa_full	delta=0	0.0000	0.0200	0.9663
hsa_full	theta_0=0	0.1000	0.2000	0.0395
hsa_full	gamma=0	0.0000	0.0200	0.9331
hsa_steady	kappa_0=0	0.1000	0.2000	2.7807
hsa_steady	delta=0	0.0000	0.0200	0.6405

8.4 Time-Varying Kappa Path



8.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-78.0366	0.5134	-177.5972	1.0000	NaN	NaN	NaN
hsa_dynamic	-62.3779	0.4434	NaN	NaN	NaN	0.0245	NaN
hsa_full	-60.0617	0.4256	NaN	NaN	0.9663	NaN	0.9331
hsa_steady	-76.7754	0.5069	-238.7793	0.0000	0.6405	NaN	NaN

9 output_gap_bn: prior=baseline, period=post_2008

Condition. data=output_gap_bn, prior=baseline, period=post_2008, constraint=unrestricted (unrestricted).

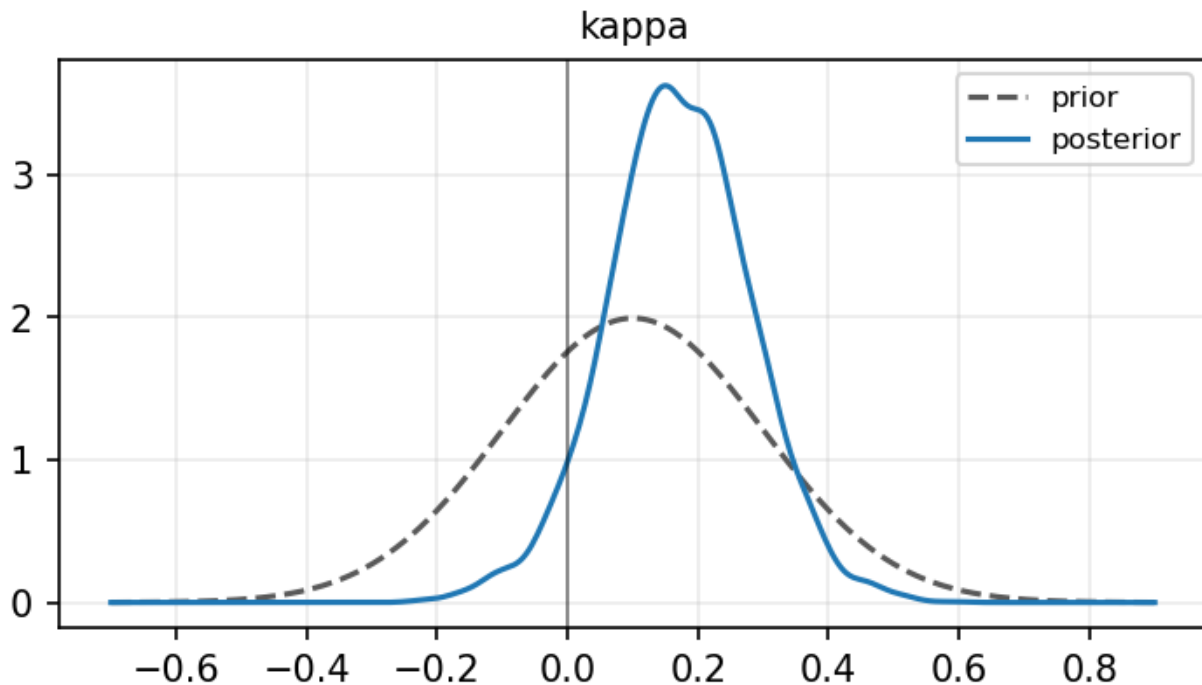
9.1 Coefficient Table

	ces	hsa_steady	hsa_dynamic	hsa_full
parameter				
alpha	0.534 [0.288, 0.776]	0.526 [0.353, 0.718]	0.550 [0.327, 0.768]	0.536 [0.287, 0.767]
kappa	0.171 [-0.050, 0.384]	NaN	0.155 [-0.046, 0.356]	NaN
kappa_0	NaN	0.135 [-0.271, 0.613]	NaN	0.169 [-0.041, 0.390]
delta	NaN	0.000 [-0.002, 0.003]	NaN	0.001 [-0.038, 0.041]
theta	NaN	NaN	0.120 [-0.306, 0.551]	NaN
theta_0	NaN	NaN	NaN	0.091 [-0.289, 0.482]
gamma	NaN	NaN	NaN	0.000 [-0.039, 0.039]
rho_1	NaN	0.505 [0.199, 0.858]	0.521 [0.129, 0.906]	0.575 [0.178, 0.962]
rho_2	NaN	-0.448 [-0.657, -0.127]	-0.355 [-0.734, -0.007]	-0.361 [-0.730, 0.013]
phi_1	0.842 [0.720, 0.961]	0.855 [0.714, 0.950]	0.845 [0.747, 0.940]	0.842 [0.717, 0.958]
lambda_ez	0.258 [-0.411, 0.915]	0.180 [-0.422, 0.685]	NaN	0.253 [-0.431, 0.907]
n	NaN	-0.069 [-0.212, 0.189]	-0.014 [-0.164, 0.145]	-0.002 [-0.173, 0.173]

9.2 Prior vs Posterior by Model

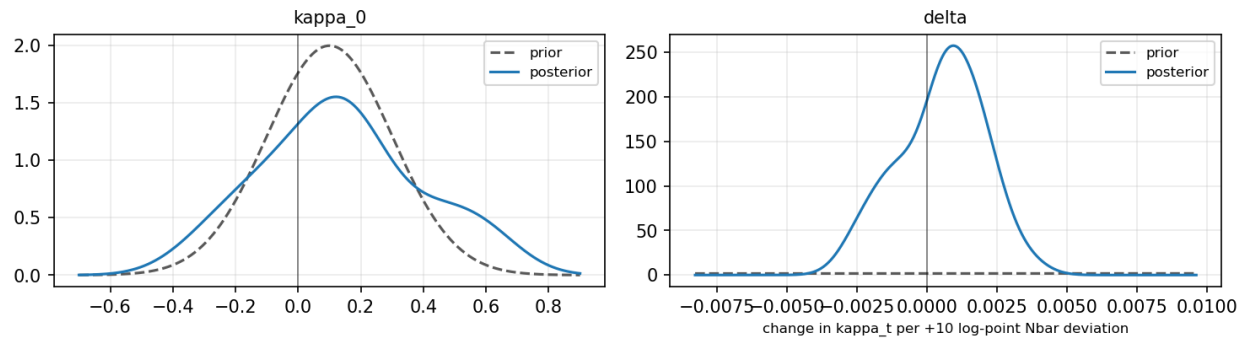
ces

Prior vs posterior — ces



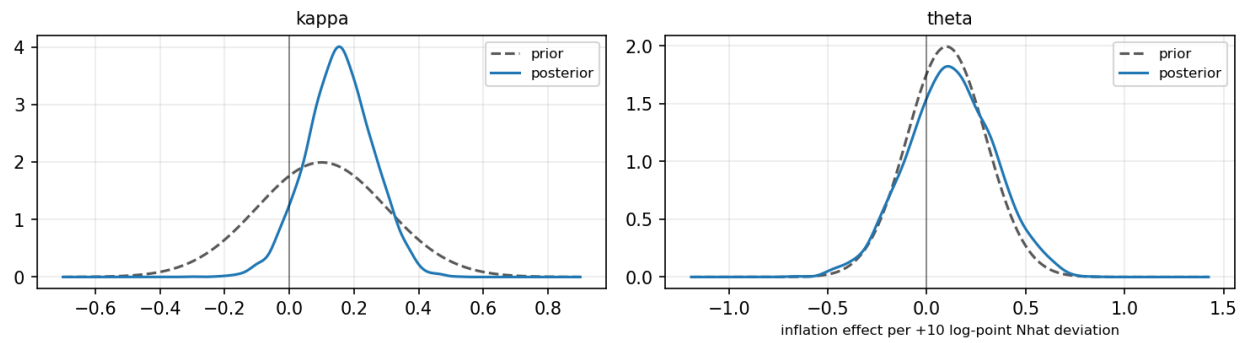
hsa_steady

Prior vs posterior — hsa_steady



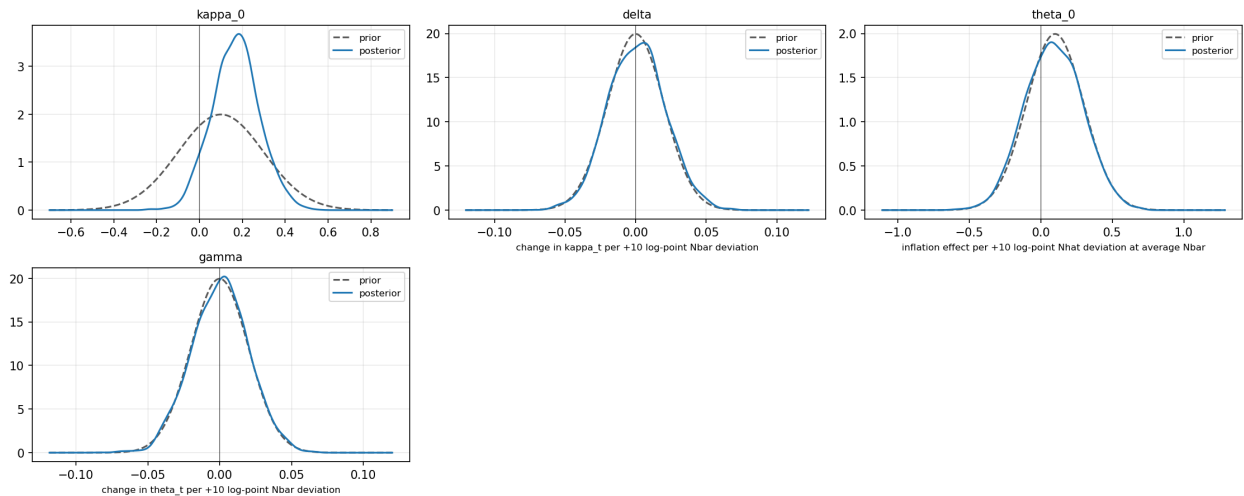
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

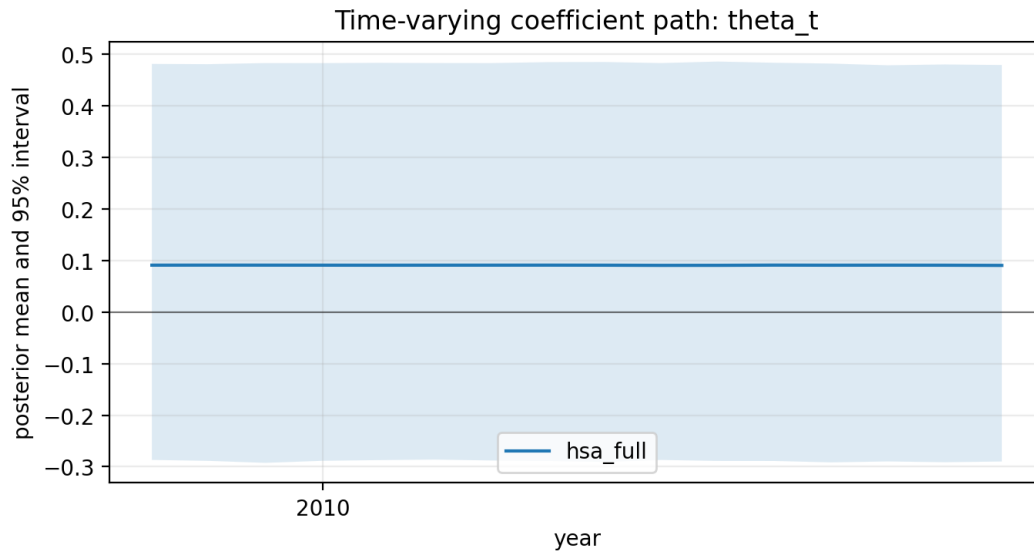
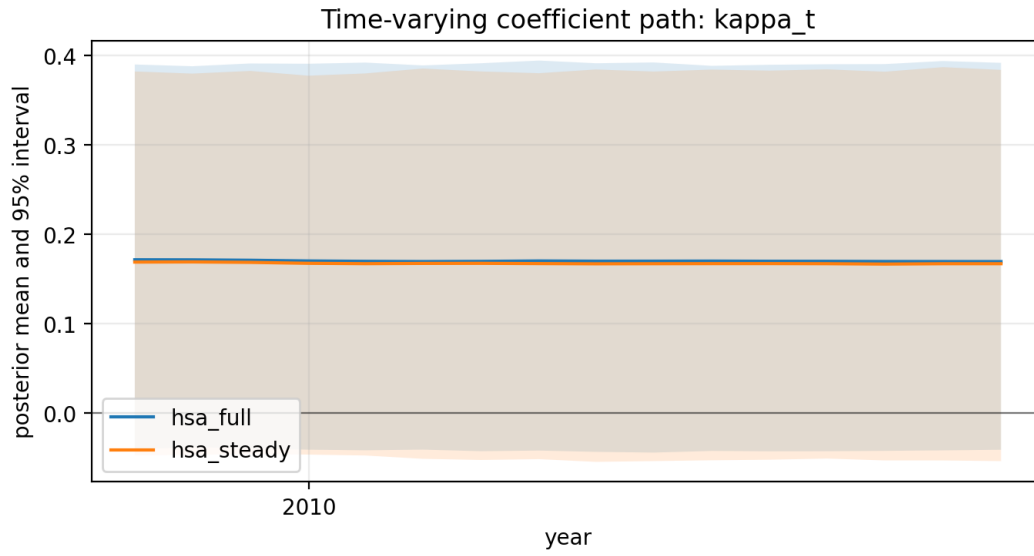
Prior vs posterior — hsa_full



9.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	0.5622
hsa_dynamic	kappa=0	0.1000	0.2000	0.7106
hsa_dynamic	theta=0	0.1000	0.2000	0.8791
hsa_full	kappa_0=0	0.1000	0.2000	0.6756
hsa_full	delta=0	0.0000	0.0200	0.9234
hsa_full	theta_0=0	0.1000	0.2000	0.9842
hsa_full	gamma=0	0.0000	0.0200	0.9914
hsa_steady	kappa_0=0	0.1000	0.2000	0.5915
hsa_steady	delta=0	0.0000	0.0200	0.9456

9.4 Time-Varying Kappa Path



9.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-20.0646	0.8322	-43.3095	1.0000	NaN	NaN	NaN
hsa_dynamic	-19.7445	0.8309	NaN	NaN	NaN	0.8791	NaN
hsa_full	-19.9845	0.8296	NaN	NaN	0.9234	NaN	0.9914
hsa_steady	-20.0744	0.8319	-66.8506	0.0000	0.9456	NaN	NaN

10 output_gap_bn: prior=baseline, period=exclude_covid

Condition. data=output_gap_bn, prior=baseline, period=exclude_covid, constraint=unrestricted (unrestricted).

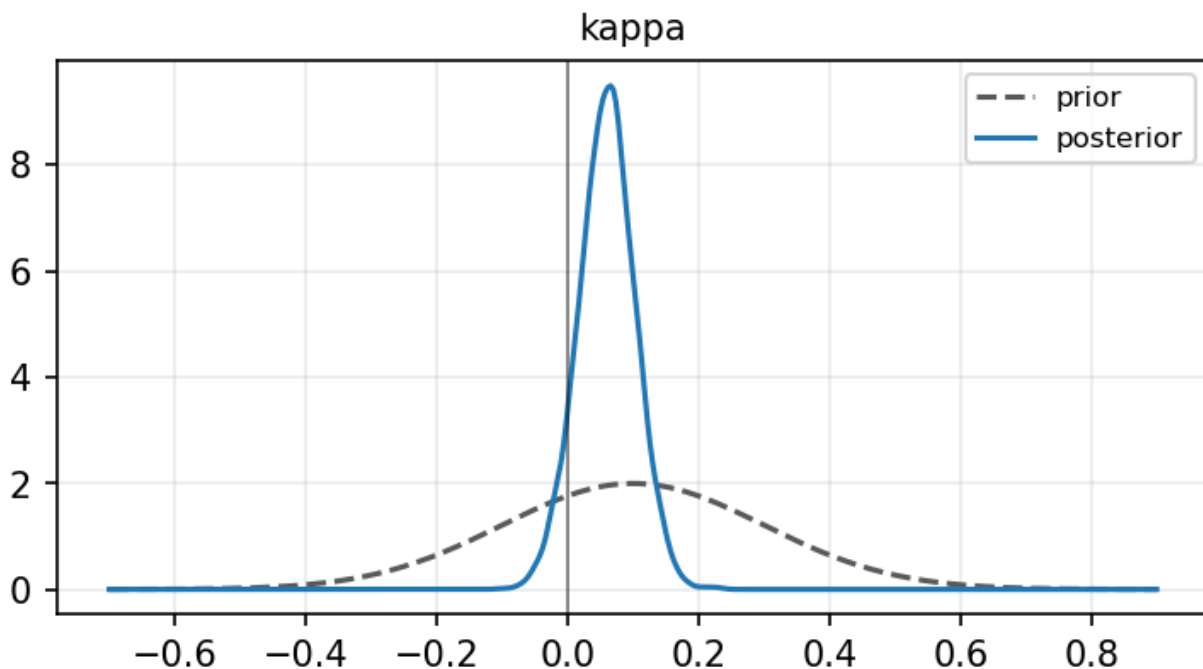
10.1 Coefficient Table

	ces	hsa_steady	hsa_dynamic	hsa_full
parameter				
alpha	0.671 [0.552, 0.787]	0.672 [0.563, 0.749]	0.629 [0.508, 0.750]	0.614 [0.496, 0.730]
kappa	0.060 [-0.025, 0.143]	NaN	0.091 [-0.002, 0.185]	NaN
kappa_0	NaN	0.157 [-0.110, 0.486]	NaN	0.096 [0.000, 0.191]
delta	NaN	-0.001 [-0.002, 0.001]	NaN	-0.007 [-0.046, 0.032]
theta	NaN	NaN	0.212 [-0.303, 0.668]	NaN
theta_0	NaN	NaN	NaN	0.233 [0.066, 0.512]
gamma	NaN	NaN	NaN	-0.010 [-0.046, 0.030]
rho_1	NaN	1.828 [1.735, 1.895]	0.920 [0.684, 1.125]	1.001 [0.769, 1.216]
rho_2	NaN	-0.832 [-0.898, -0.740]	0.016 [-0.183, 0.228]	-0.039 [-0.250, 0.185]
phi_1	0.915 [0.850, 0.979]	0.910 [0.834, 0.973]	0.915 [0.853, 0.979]	0.915 [0.850, 0.979]
lambda_ez	0.009 [-0.223, 0.231]	-0.028 [-0.260, 0.244]	NaN	-0.023 [-0.248, 0.199]
n	NaN	-0.070 [-0.188, 0.092]	-0.019 [-0.076, 0.029]	0.005 [-0.049, 0.062]

10.2 Prior vs Posterior by Model

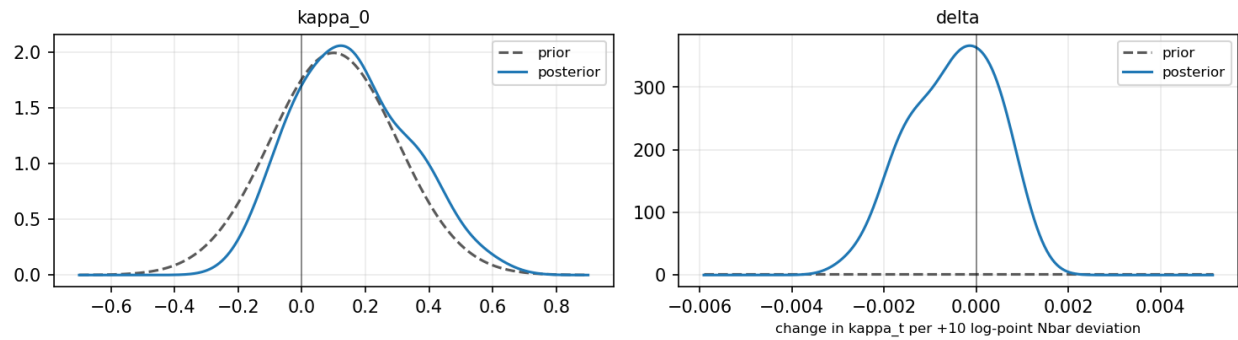
ces

Prior vs posterior — ces



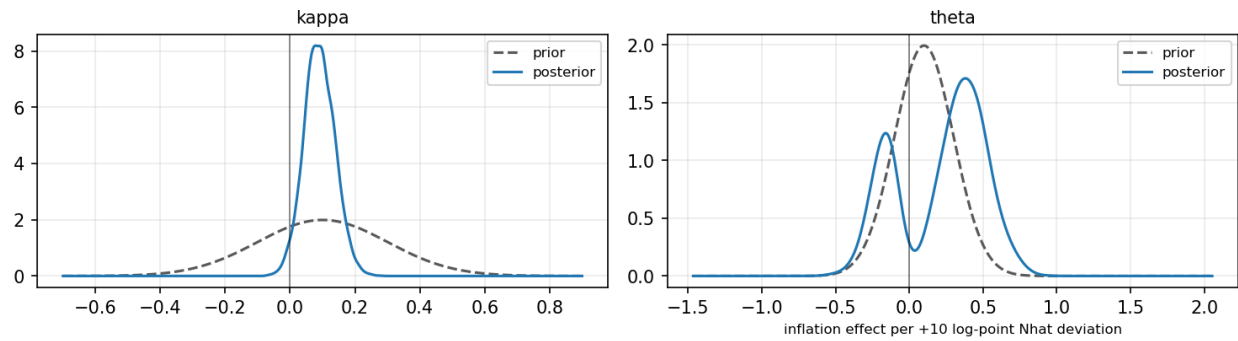
hsa_steady

Prior vs posterior — hsa_steady



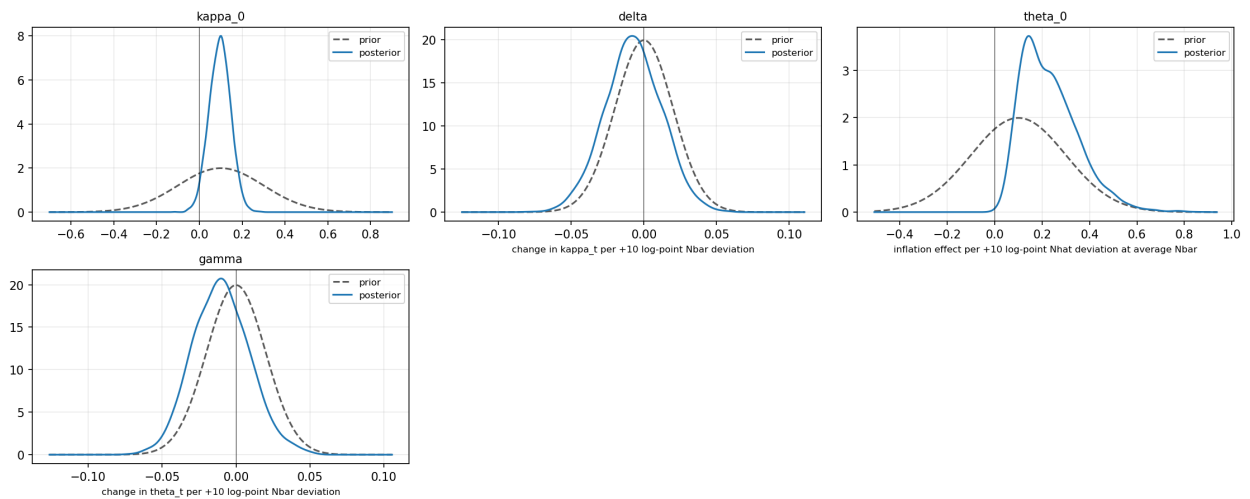
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

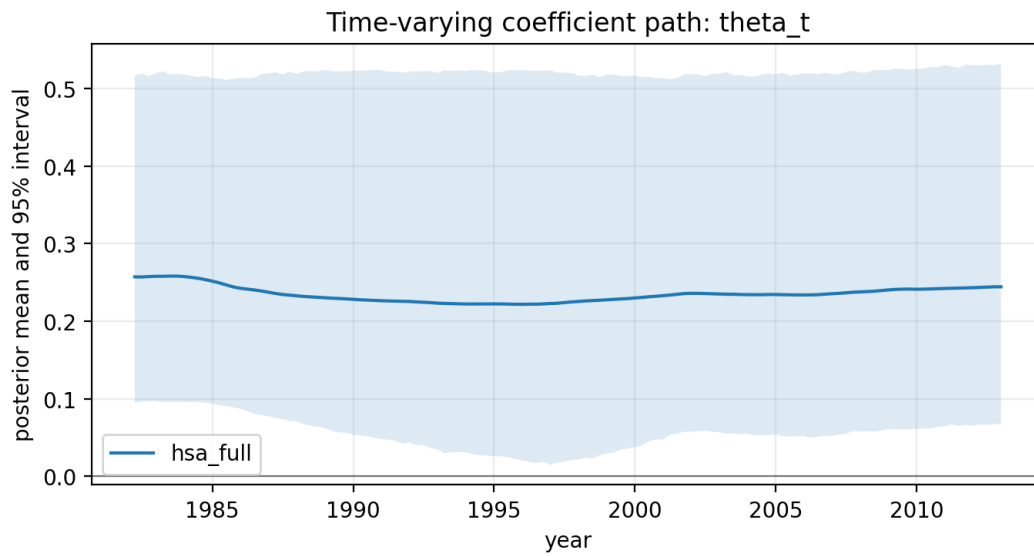
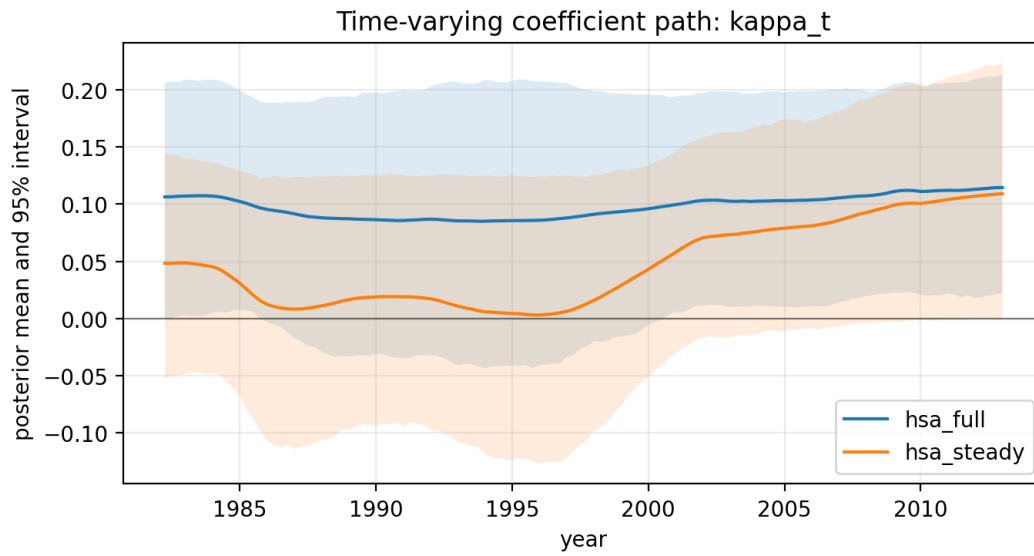
Prior vs posterior — hsa_full



10.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	1.9113
hsa_dynamic	kappa=0	0.1000	0.2000	0.7598
hsa_dynamic	theta=0	0.1000	0.2000	0.1635
hsa_full	kappa_0=0	0.1000	0.2000	0.7535
hsa_full	delta=0	0.0000	0.0200	0.9277
hsa_full	theta_0=0	0.1000	0.2000	0.0439
hsa_full	gamma=0	0.0000	0.0200	0.8528
hsa_steady	kappa_0=0	0.1000	0.2000	2.9299
hsa_steady	delta=0	0.0000	0.0200	0.4640

10.4 Time-Varying Kappa Path



10.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-120.7912	0.6396	-245.3535	1.0000	NaN	NaN	NaN
hsa_dynamic	-118.2796	0.6267	NaN	NaN	NaN	0.1635	NaN
hsa_full	-107.8056	0.5744	NaN	NaN	0.9277	NaN	0.8528
hsa_steady	-118.9122	0.6299	-306.6561	0.0000	0.4640	NaN	NaN

11 output_gap_bn: prior=baseline, period=start_1988

Condition. data=output_gap_bn, prior=baseline, period=start_1988, constraint=unrestricted (unrestricted).

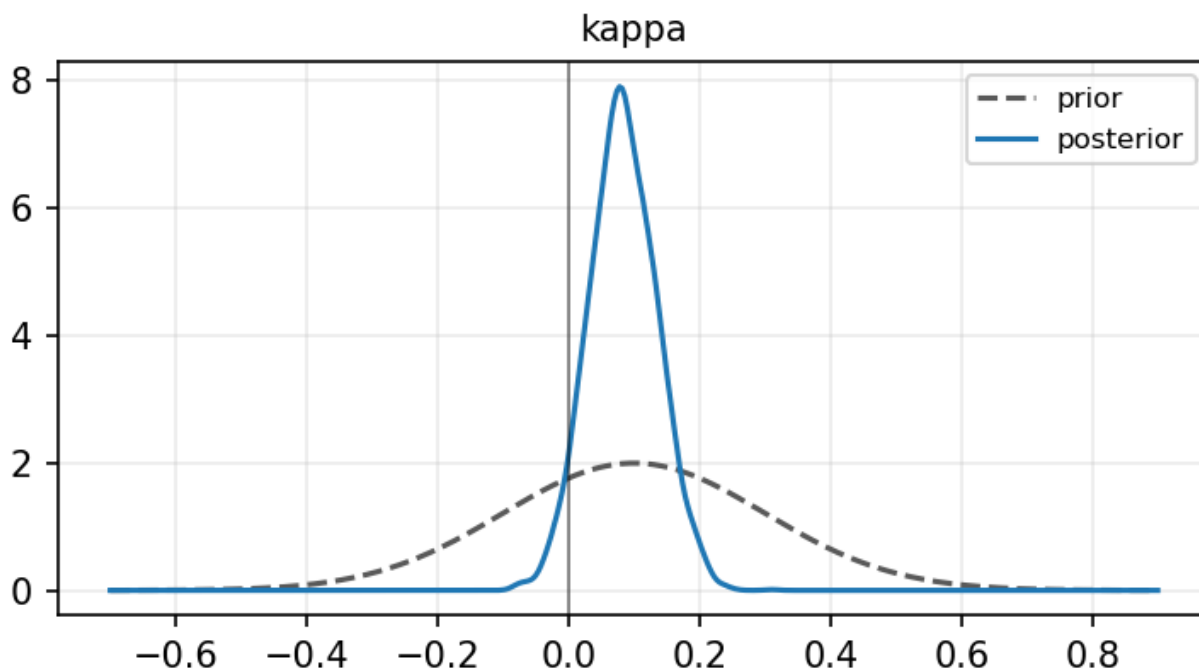
11.1 Coefficient Table

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.641 [0.515, 0.774]	0.642 [0.523, 0.777]	0.602 [0.466, 0.734]	0.570 [0.432, 0.710]
kappa	0.083 [-0.016, 0.186]	NaN	0.131 [0.010, 0.259]	NaN
kappa_0	NaN	0.097 [-0.281, 0.461]	NaN	0.140 [0.018, 0.274]
delta	NaN	-0.000 [-0.002, 0.002]	NaN	-0.019 [-0.054, 0.016]
theta	NaN	NaN	0.276 [-0.159, 0.630]	NaN
theta_0	NaN	NaN	NaN	0.284 [0.051, 0.531]
gamma	NaN	NaN	NaN	-0.005 [-0.044, 0.033]
rho_1	NaN	1.744 [1.636, 1.867]	0.801 [0.344, 1.099]	0.953 [0.706, 1.193]
rho_2	NaN	-0.748 [-0.875, -0.638]	-0.015 [-0.306, 0.234]	-0.033 [-0.269, 0.194]
phi_1	0.912 [0.837, 0.986]	0.912 [0.849, 0.982]	0.913 [0.837, 0.992]	0.912 [0.837, 0.984]
lambda_ez	-0.024 [-0.298, 0.250]	0.007 [-0.243, 0.249]	NaN	-0.077 [-0.360, 0.198]
n	NaN	-0.212 [-0.365, -0.003]	-0.040 [-0.082, 0.003]	-0.039 [-0.098, 0.022]

11.2 Prior vs Posterior by Model

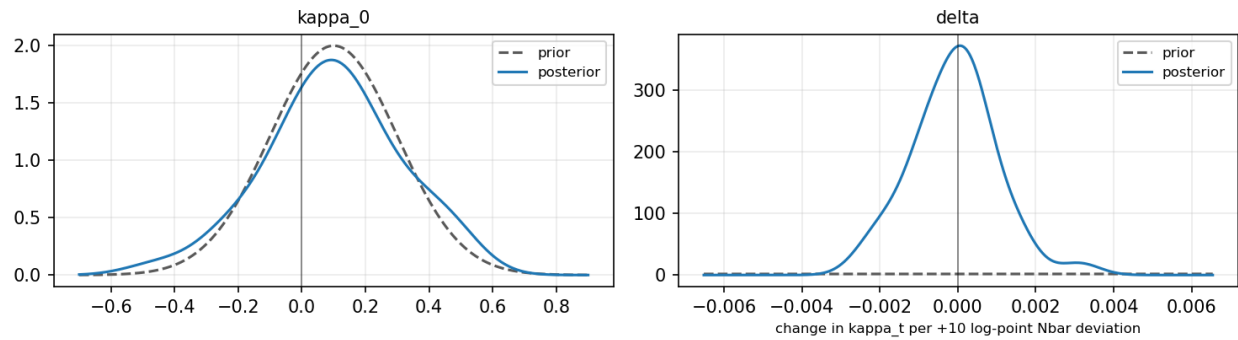
ces

Prior vs posterior — ces



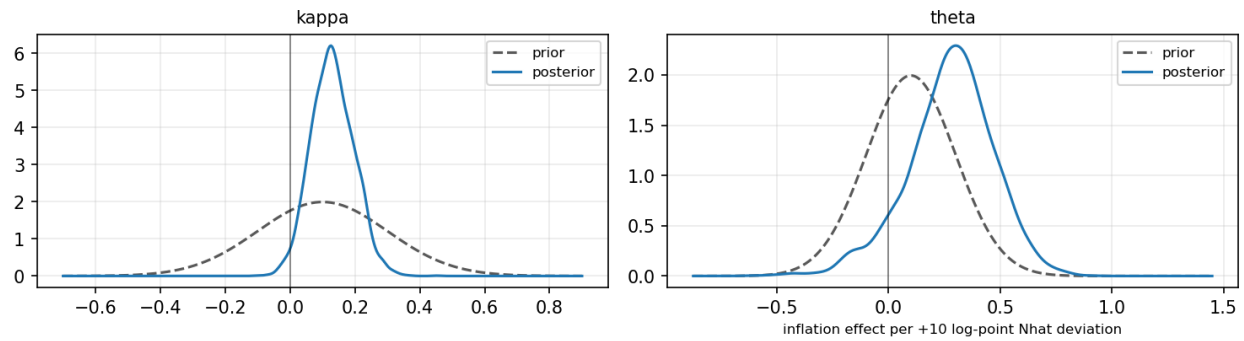
hsa_steady

Prior vs posterior — hsa_steady



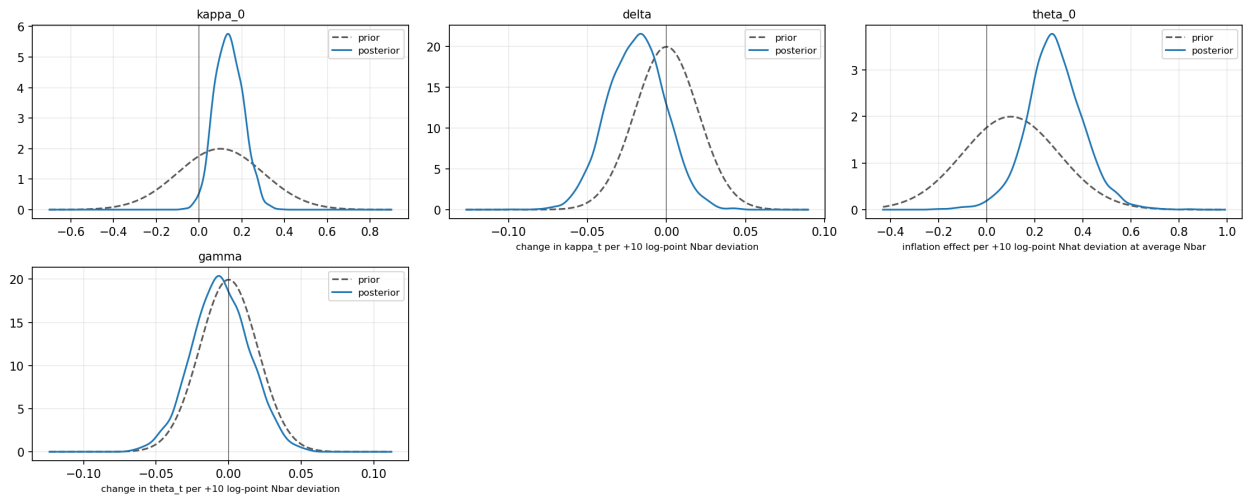
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

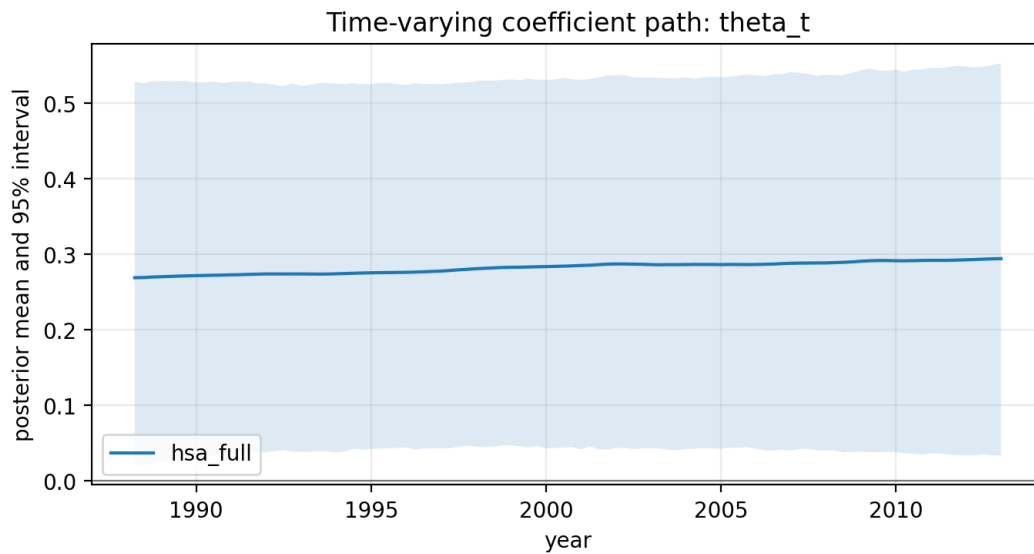
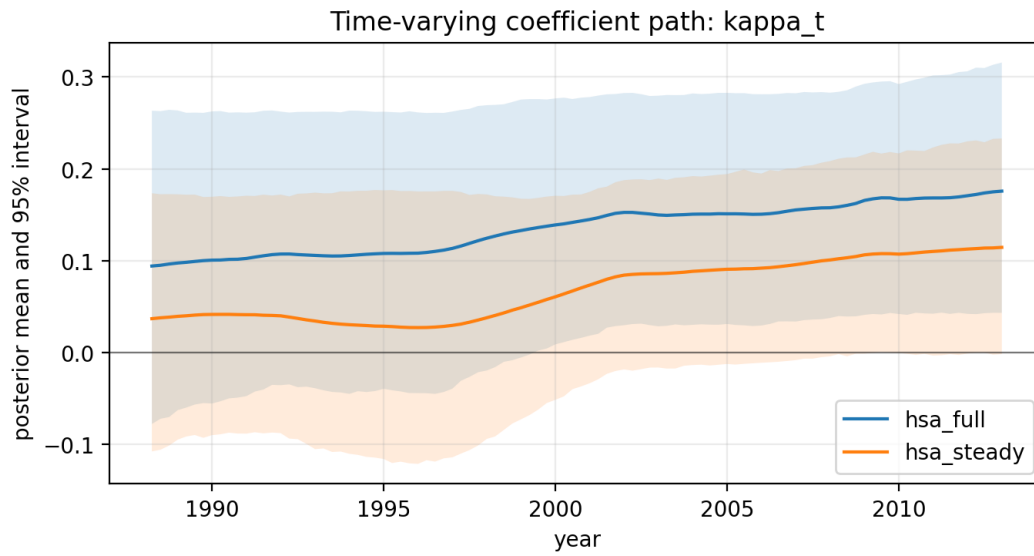
Prior vs posterior — hsa_full



11.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	1.2030
hsa_dynamic	kappa=0	0.1000	0.2000	0.4174
hsa_dynamic	theta=0	0.1000	0.2000	0.3488
hsa_full	kappa_0=0	0.1000	0.2000	0.3091
hsa_full	delta=0	0.0000	0.0200	0.6430
hsa_full	theta_0=0	0.1000	0.2000	0.1086
hsa_full	gamma=0	0.0000	0.0200	0.9280
hsa_steady	kappa_0=0	0.1000	0.2000	1.8892
hsa_steady	delta=0	0.0000	0.0200	0.6282

11.4 Time-Varying Kappa Path



11.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-93.4568	0.6140	-185.8159	1.0000	NaN	NaN	NaN
hsa_dynamic	-87.7238	0.5815	NaN	NaN	NaN	0.3488	NaN
hsa_full	-83.8885	0.5552	NaN	NaN	0.6430	NaN	0.9280
hsa_steady	-92.2986	0.6069	-244.5849	0.0000	0.6282	NaN	NaN

12 output_gap_bn: prior=baseline, period=end_2019

Condition. data=output_gap_bn, prior=baseline, period=end_2019, constraint=unrestricted (unrestricted).

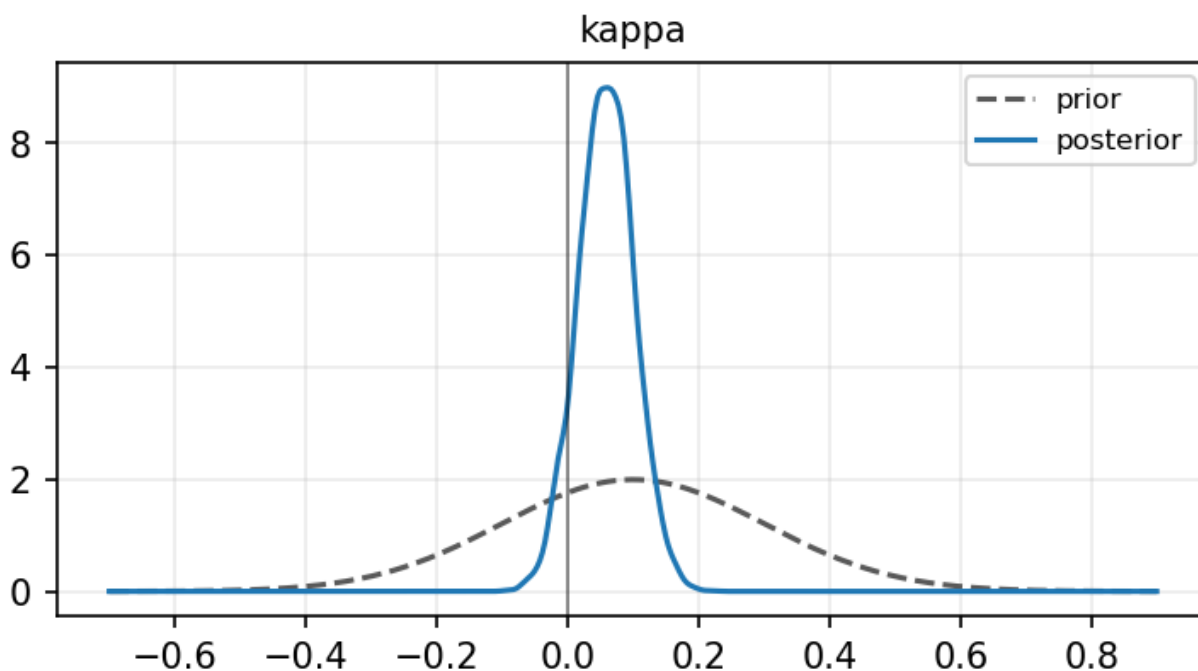
12.1 Coefficient Table

	ces	hsa_steady	hsa_dynamic	hsa_full
parameter				
alpha	0.672 [0.554, 0.785]	0.681 [0.570, 0.766]	0.629 [0.508, 0.747]	0.618 [0.490, 0.738]
kappa	0.059 [-0.024, 0.142]	NaN	0.091 [-0.000, 0.186]	NaN
kappa_0	NaN	0.110 [-0.145, 0.474]	NaN	0.083 [-0.013, 0.185]
delta	NaN	-0.000 [-0.002, 0.001]	NaN	-0.020 [-0.052, 0.014]
theta	NaN	NaN	0.193 [-0.326, 0.656]	NaN
theta_0	NaN	NaN	NaN	0.093 [-0.333, 0.585]
gamma	NaN	NaN	NaN	-0.003 [-0.043, 0.035]
rho_1	NaN	1.808 [1.740, 1.866]	0.913 [0.663, 1.126]	0.964 [0.732, 1.200]
rho_2	NaN	-0.812 [-0.869, -0.742]	0.016 [-0.191, 0.234]	-0.030 [-0.251, 0.192]
phi_1	0.915 [0.850, 0.977]	0.917 [0.871, 0.972]	0.915 [0.855, 0.977]	0.914 [0.848, 0.980]
lambda_ez	0.009 [-0.224, 0.242]	0.005 [-0.245, 0.139]	NaN	-0.014 [-0.244, 0.221]
n	NaN	-0.030 [-0.148, 0.147]	-0.020 [-0.076, 0.031]	-0.027 [-0.093, 0.037]

12.2 Prior vs Posterior by Model

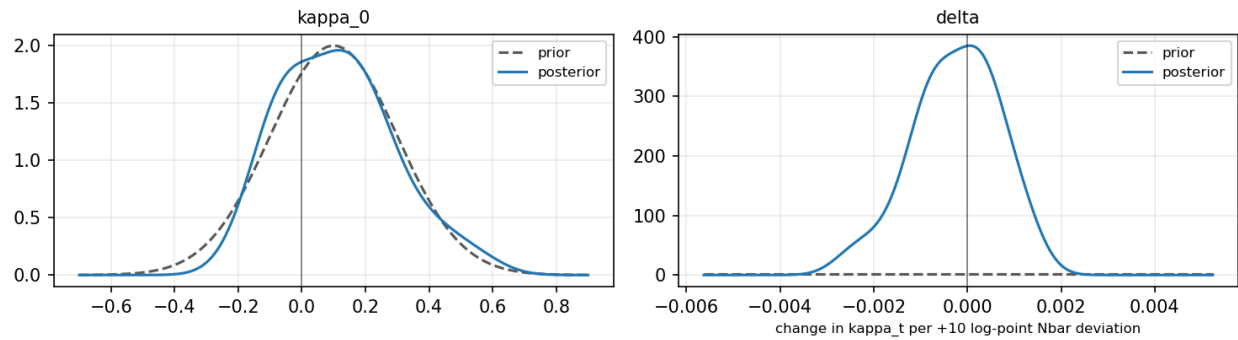
ces

Prior vs posterior — ces



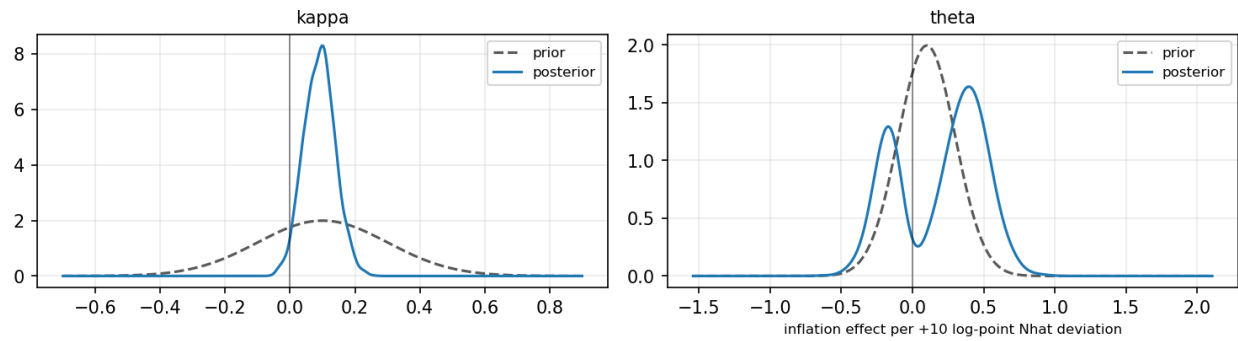
hsa_steady

Prior vs posterior — hsa_steady



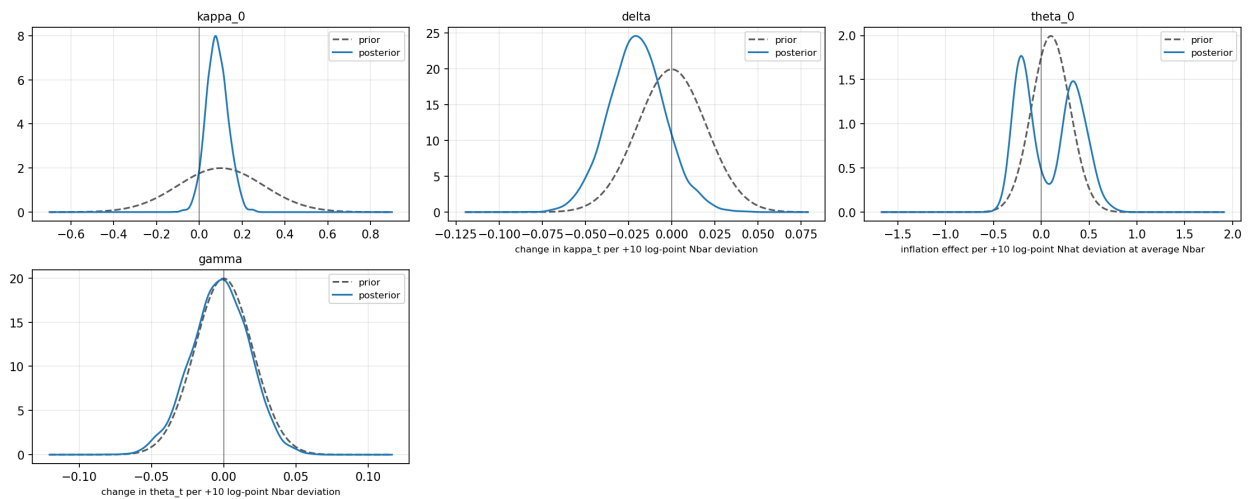
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

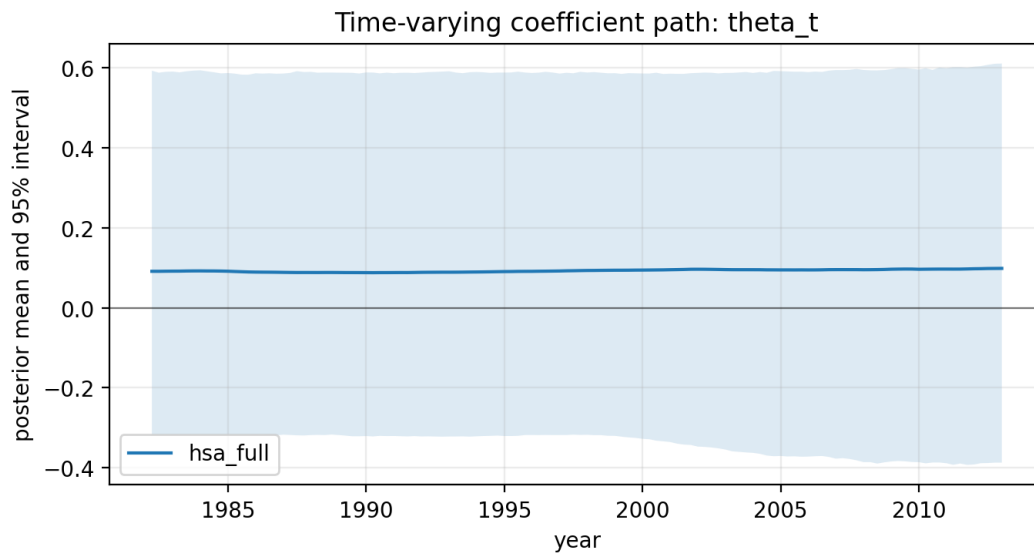
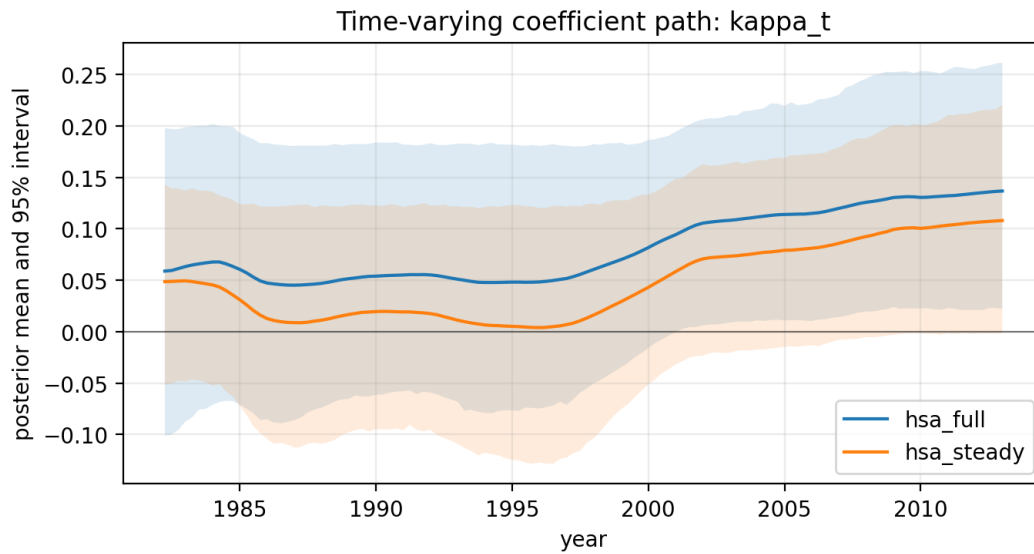
Prior vs posterior — hsa_full



12.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	1.8850
hsa_dynamic	kappa=0	0.1000	0.2000	0.7983
hsa_dynamic	theta=0	0.1000	0.2000	0.1814
hsa_full	kappa_0=0	0.1000	0.2000	1.1031
hsa_full	delta=0	0.0000	0.0200	0.5398
hsa_full	theta_0=0	0.1000	0.2000	0.2718
hsa_full	gamma=0	0.0000	0.0200	0.9929
hsa_steady	kappa_0=0	0.1000	0.2000	2.8353
hsa_steady	delta=0	0.0000	0.0200	0.4987

12.4 Time-Varying Kappa Path



12.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-120.7953	0.6396	-245.3487	1.0000	NaN	NaN	NaN
hsa_dynamic	-119.1190	0.6308	NaN	NaN	NaN	0.1814	NaN
hsa_full	-121.1207	0.6414	NaN	NaN	0.5398	NaN	0.9929
hsa_steady	-118.8731	0.6298	-306.5956	0.0000	0.4987	NaN	NaN

13 output_gap_bn: prior=weak, period=full

Condition. data=output_gap_bn, prior=weak, period=full, constraint=unrestricted (unrestricted).

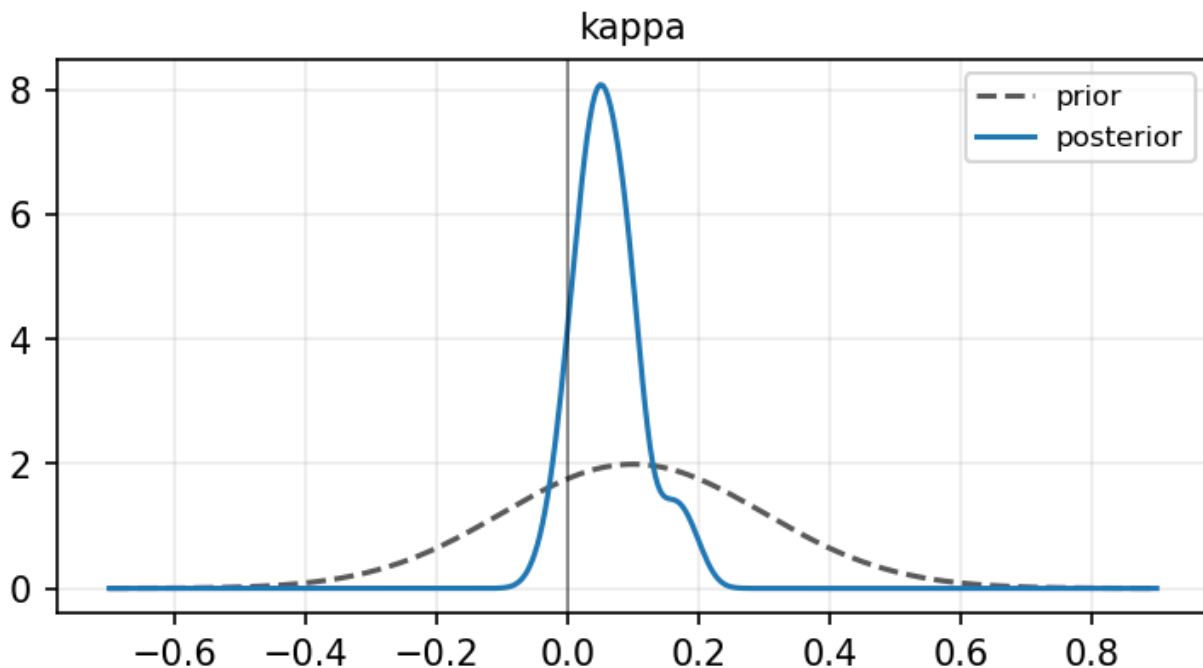
13.1 Coefficient Table

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.669 [0.544, 0.784]	0.655 [0.534, 0.754]	0.642 [0.541, 0.730]	0.662 [0.563, 0.758]
kappa	0.062 [-0.006, 0.176]	NaN	0.081 [-0.004, 0.150]	NaN
kappa_0	NaN	0.136 [-0.243, 0.429]	NaN	0.248 [-0.039, 0.547]
delta	NaN	-0.000 [-0.002, 0.002]	NaN	-0.001 [-0.002, 0.001]
theta	NaN	NaN	0.009 [0.001, 0.014]	NaN
theta_0	NaN	NaN	NaN	0.101 [-0.175, 0.409]
gamma	NaN	NaN	NaN	-0.000 [-0.002, 0.001]
rho_1	NaN	1.802 [1.700, 1.893]	1.862 [1.794, 1.913]	1.191 [1.003, 1.388]
rho_2	NaN	-0.806 [-0.900, -0.703]	-0.866 [-0.917, -0.797]	-0.269 [-0.470, -0.067]
phi_1	0.919 [0.838, 0.970]	0.918 [0.867, 0.974]	0.907 [0.861, 0.953]	0.916 [0.872, 0.976]
lambda_ez	-0.009 [-0.205, 0.221]	0.001 [-0.161, 0.161]	NaN	-0.013 [-0.195, 0.272]
n	NaN	-0.057 [-0.200, 0.059]	-0.026 [-0.150, 0.106]	-0.090 [-0.181, 0.037]

13.2 Prior vs Posterior by Model

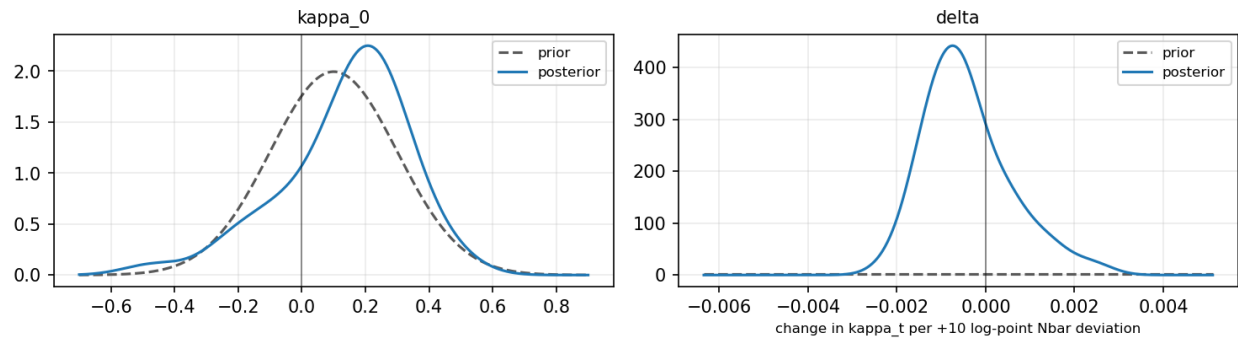
ces

Prior vs posterior — ces



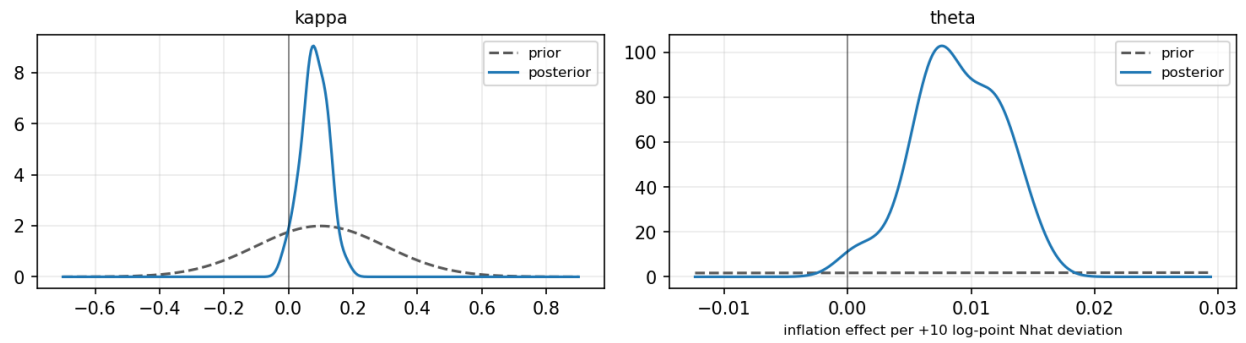
hsa_steady

Prior vs posterior — hsa_steady



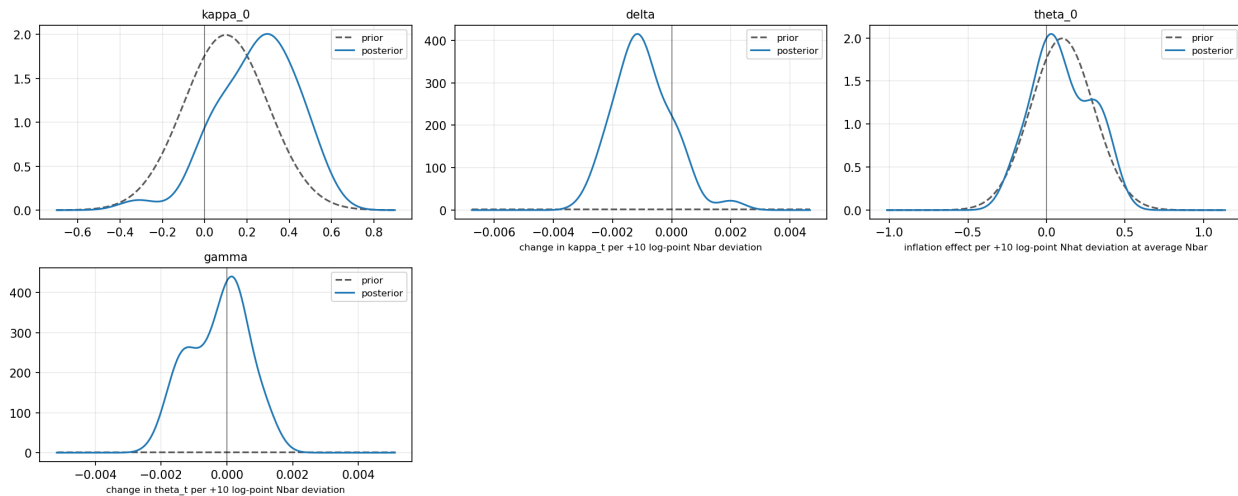
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

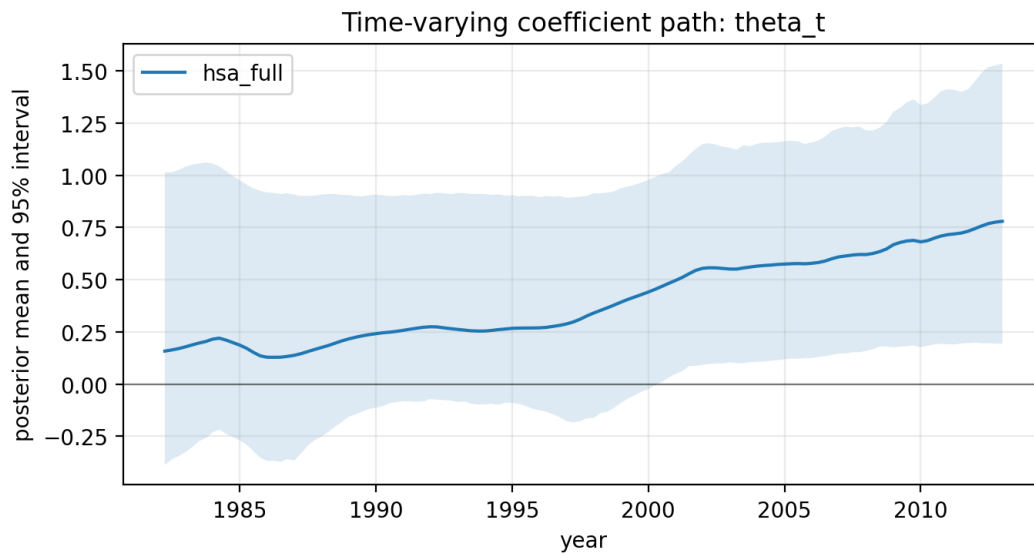
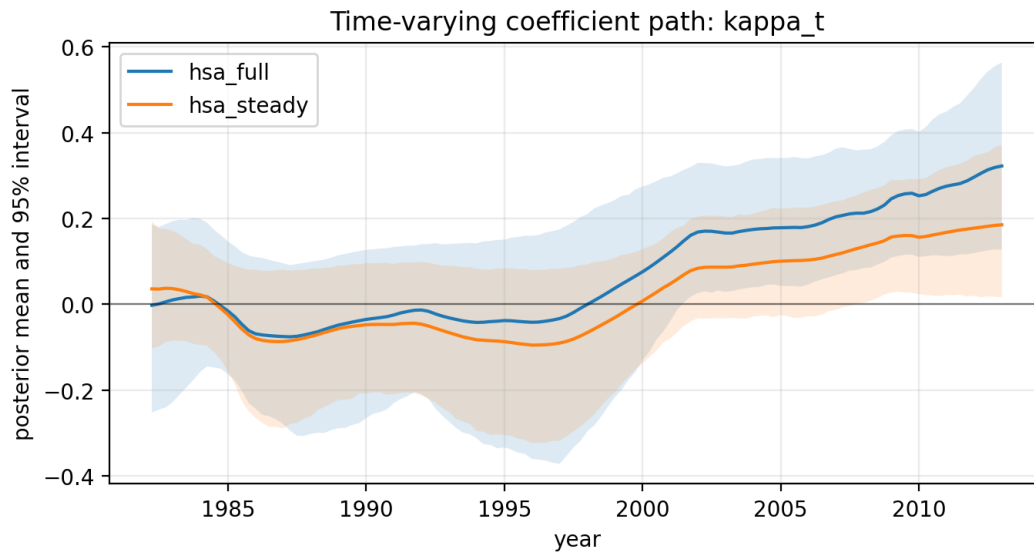
Prior vs posterior — hsa_full



13.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.5000	4.1549
hsa_dynamic	kappa=0	0.1000	0.5000	1.2848
hsa_dynamic	theta=0	0.0000	0.5000	0.1700
hsa_full	kappa_0=0	0.1000	0.5000	3.6211
hsa_full	delta=0	0.0000	0.1000	0.0947
hsa_full	theta_0=0	0.0000	0.5000	0.1603
hsa_full	gamma=0	0.0000	0.1000	0.3917
hsa_steady	kappa_0=0	0.1000	0.5000	6.1808
hsa_steady	delta=0	0.0000	0.1000	0.3615

13.4 Time-Varying Kappa Path



13.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-120.6571	0.6395	-245.3920	1.0000	NaN	NaN	NaN
hsa_dynamic	-118.0198	0.6253	NaN	NaN	NaN	0.1700	NaN
hsa_full	-116.1073	0.6156	NaN	NaN	0.0947	NaN	0.3917
hsa_steady	-116.5241	0.6181	-297.6572	0.0000	0.3615	NaN	NaN

14 output_gap_bn: prior=tight, period=full

Condition. data=output_gap_bn, prior=tight, period=full, constraint=unrestricted (unrestricted).

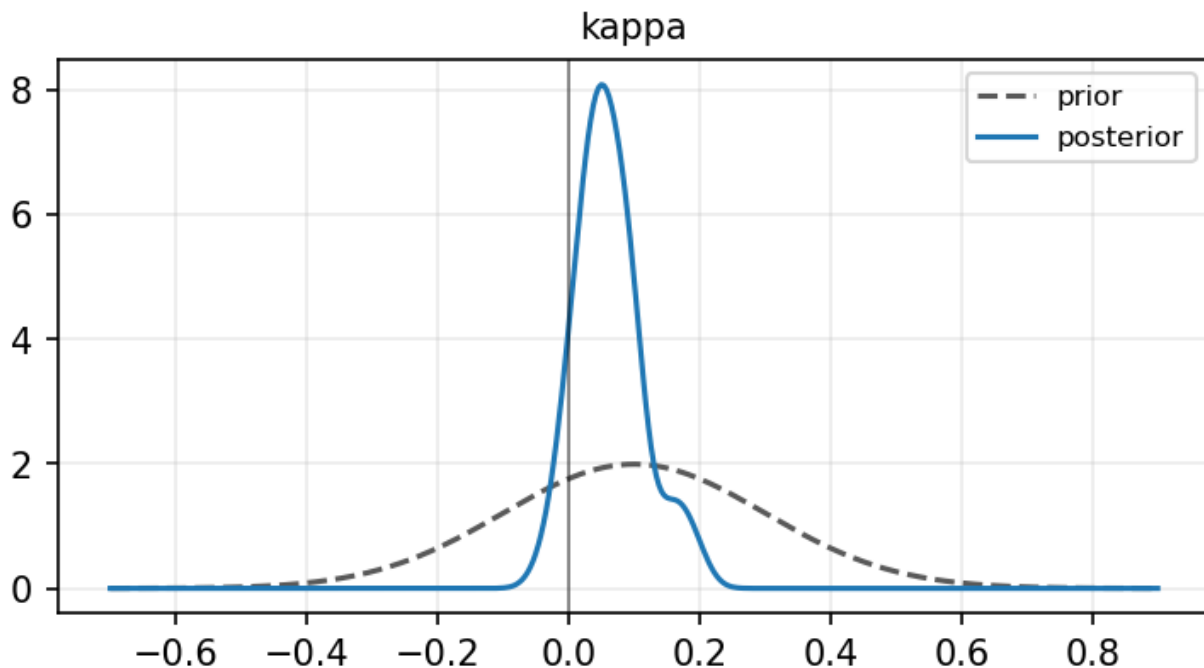
14.1 Coefficient Table

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.669 [0.544, 0.784]	0.655 [0.534, 0.754]	0.642 [0.541, 0.730]	0.662 [0.563, 0.758]
kappa	0.062 [-0.006, 0.176]	NaN	0.081 [-0.004, 0.150]	NaN
kappa_0	NaN	0.136 [-0.243, 0.429]	NaN	0.248 [-0.039, 0.547]
delta	NaN	-0.000 [-0.002, 0.002]	NaN	-0.001 [-0.002, 0.001]
theta	NaN	NaN	0.009 [0.001, 0.014]	NaN
theta_0	NaN	NaN	NaN	0.101 [-0.175, 0.409]
gamma	NaN	NaN	NaN	-0.000 [-0.002, 0.001]
rho_1	NaN	1.802 [1.700, 1.893]	1.862 [1.794, 1.913]	1.191 [1.003, 1.388]
rho_2	NaN	-0.806 [-0.900, -0.703]	-0.866 [-0.917, -0.797]	-0.269 [-0.470, -0.067]
phi_1	0.919 [0.838, 0.970]	0.918 [0.867, 0.974]	0.907 [0.861, 0.953]	0.916 [0.872, 0.976]
lambda_ez	-0.009 [-0.205, 0.221]	0.001 [-0.161, 0.161]	NaN	-0.013 [-0.195, 0.272]
n	NaN	-0.057 [-0.200, 0.059]	-0.026 [-0.150, 0.106]	-0.090 [-0.181, 0.037]

14.2 Prior vs Posterior by Model

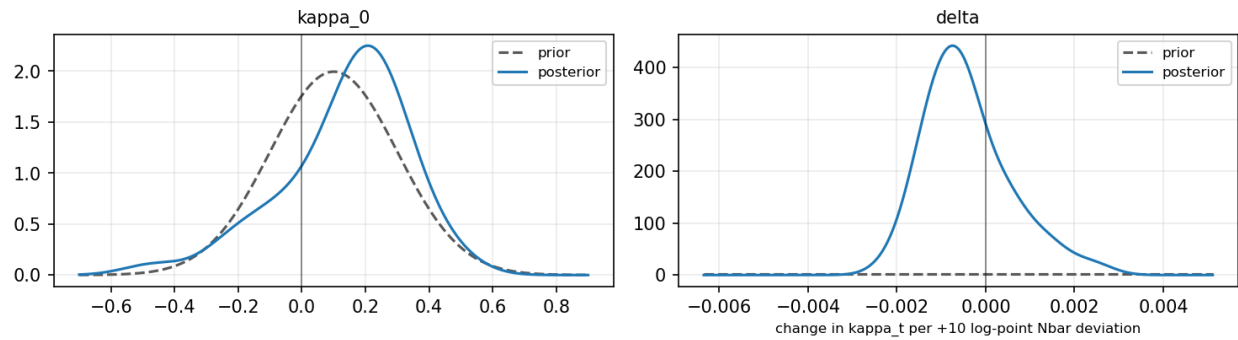
ces

Prior vs posterior — ces



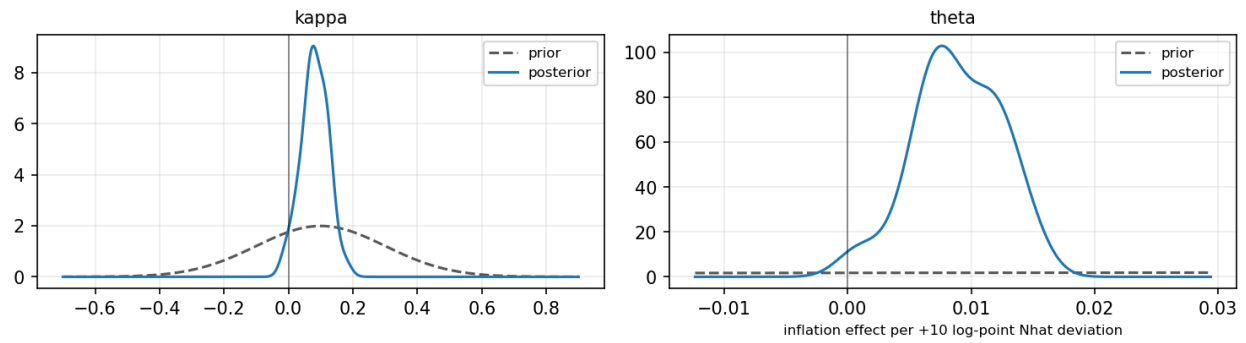
hsa_steady

Prior vs posterior — hsa_steady



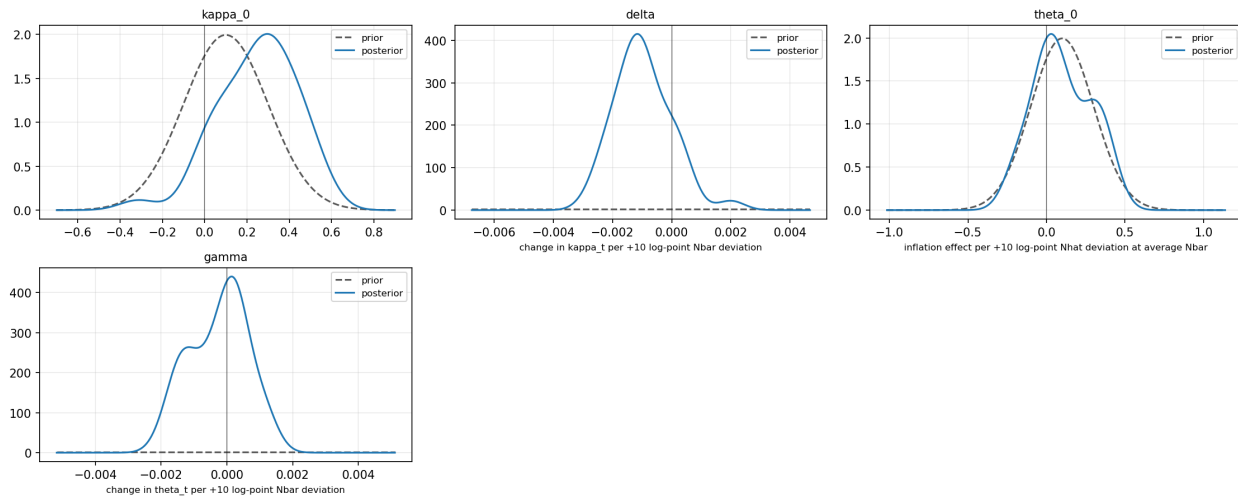
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

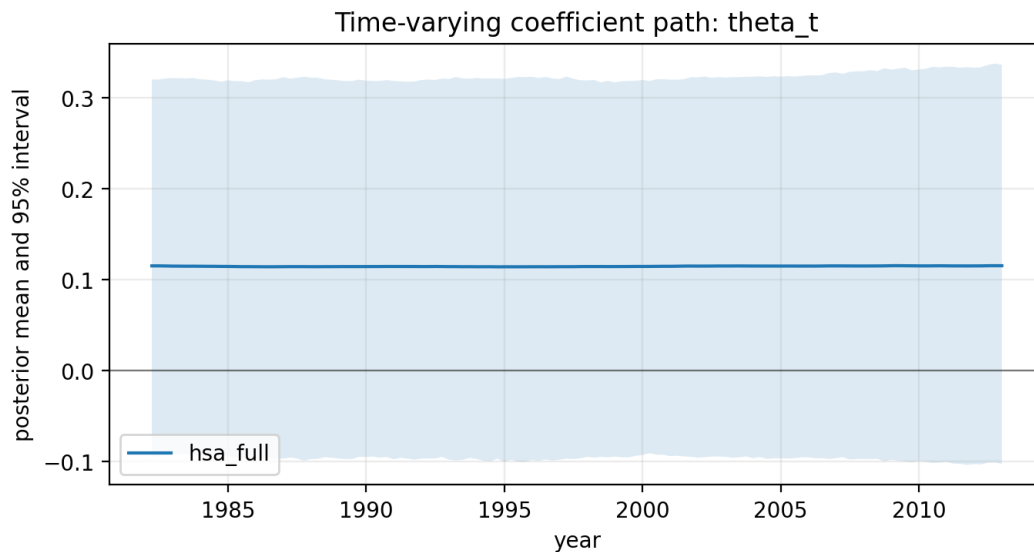
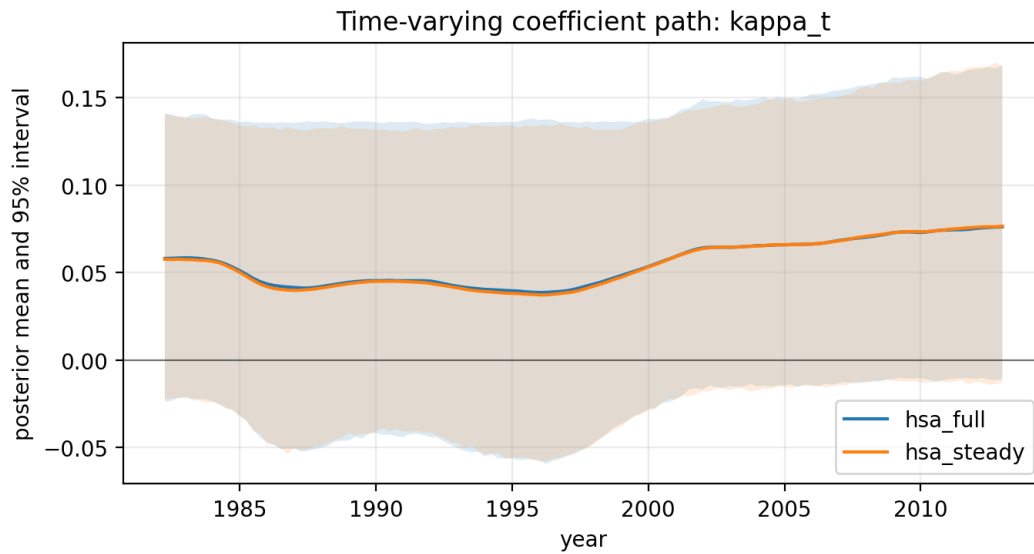
Prior vs posterior — hsa_full



14.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.1000	1.5005
hsa_dynamic	kappa=0	0.1000	0.1000	1.2251
hsa_dynamic	theta=0	0.1000	0.1000	0.9551
hsa_full	kappa_0=0	0.1000	0.1000	1.6920
hsa_full	delta=0	0.0000	0.0100	0.7170
hsa_full	theta_0=0	0.1000	0.1000	0.8922
hsa_full	gamma=0	0.0000	0.0100	1.0356
hsa_steady	kappa_0=0	0.1000	0.1000	1.7016
hsa_steady	delta=0	0.0000	0.0100	0.7211

14.4 Time-Varying Kappa Path



14.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-121.4013	0.6416	-245.3446	1.0000	NaN	NaN	NaN
hsa_dynamic	-120.8729	0.6411	NaN	NaN	NaN	0.9551	NaN
hsa_full	-120.0090	0.6344	NaN	NaN	0.7170	NaN	1.0356
hsa_steady	-120.6256	0.6375	-332.1100	0.0000	0.7211	NaN	NaN

15 Notes

WAIC and LOO are not used as primary criteria because the models contain latent states. Chib marginal likelihoods are reported only when the legacy ordinate calculation can be matched to the sampler units and supplied data.